

SPECIFICATION
of
ELECTRICAL WORK
for
Kitimat Fish Hatchery Electrical Service Upgrade
Kitimat, BC
for
Department of Fisheries & Oceans

ELECTRICAL CONSULTANT

KySu Consulting Ltd.
Terrace, BC V8G 1X6

PROJECT NO: TER-2021-006
ISSUED FOR TENDER REVIEW
ISSUE DATE: 27 June, 2025

26 05 00	Common Work Results - Electrical
26 05 20	Wire and Box Connectors (0 – 1,000 V)
26 05 21.02	Wires and Cables (0 – 600 V)
26 05 22	Connectors and Terminations
26 05 28	Grounding – Secondary
26 05 29	Hangers and Supports for Electrical Systems
26 05 31	Splitters, Junction, Pull Boxes and Cabinets
26 05 32	Outlet Boxes, Conduit Boxes and Fittings
26 05 34	Conduits, Conduit Fastenings and Conduit Fittings
26 05 43.01	Installation of Cables in Trenches and In Ducts
26 28 23	Disconnect Switches – Fused and Non-Fused
26 43 13	Surge Protection Device (SPD)

1. GENERAL

1.1 Related Sections

1. The General Conditions, Supplements and Amendments shall govern this Section (read in conjunction with Instructions to Tenderers/Bidders). This section covers items common to all Electrical sections and is intended only to supplement the requirement of Division 00 and 01.
2. Reference to “Electrical Divisions” shall mean all sections of Divisions 26, 27, and 28 in the Master Format of the Canadian Master Specifications.
3. Provide materials, equipment and plant, or specified design, performance and quality, and current models with published certified ratings for which replacement parts are readily available. Provide project management and on-site supervision to undertake administration, meet schedules, ensure timely performance, ensure coordination, and establish orderly completion and the delivery of a fully commissioned installation.
4. The most stringent requirements of this and other electrical sections shall govern.
5. All work shall be in accordance with Project Drawings and specifications and their intent, complete with all necessary components, including those not normally shown or specified, but required for a complete installation.
6. Provide seismic restraints for all required equipment, lighting and conduit.
7. Connect to equipment specified in other Sections and to equipment supplied and installed by other Trade Contractors or by the Owner. Uncrate equipment, move in place and install complete; start up and test. Include all field assembly of loosely/separately packaged accessories.

1.2 References

1. Canadian Standards Association (CSA International)
 1. CSA C22.1, Canadian Electrical Code, Part 1 (current edition), Safety Standard for Electrical Installations.
 2. Comply with all electrical CSA standards and bulletins.
 3. CAN3-C235 (current edition), Preferred Voltage Levels for AC Systems, 0 to 50,000V.
2. Electrical and Electronic Manufacturer's Association of Canada (EEMAC)
 1. EEMAC 2y-1 (current editions), Light Gray Colour for Indoor Switchgear.
3. Institute of Electrical and Electronics (IEEE)/National Electrical Safety Code Product Line (NESC)
 1. IEEE SP1122- (current edition), The Authoritative Dictionary of IEEE Standards Terms, 7th Edition.

1.3 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.4 Design Requirements

1. Operating Voltage: to CAN3-C235 (current edition).

2. Motors, electric heating, control and distribution devices and equipment to operate satisfactorily at 60 Hz within normal operating limits established by above standard.
 1. Equipment to operate in extreme operating conditions established in above standard without damage to equipment.
3. Language operating requirements: provide identification nameplates and labels for controls in English.

1.5 Definitions

1. Electrical and electronic terms: unless otherwise specified or indicated, terms used in these specifications, and on drawings, are those defined by IEEE SP1122.
2. The “Engineer” or “Consultant” is defined as KySu Consulting Ltd.
3. The “Trade Contractor” is defined as the supplier of the Scope of work as defined in the Electrical Specifications sections in Division 26.
4. “Provide” is defined as “supply, install, test and commission.”
5. “Install” is defined as all work and materials necessary to place the specified item into full operation, securely fastened, and to give a presentable finished appearance. “Install” also includes all necessary connections and conductors.
6. “Delete” or “remove” is defined as to disconnect, make safe, and remove all obsolete materials back to source including but not limited to raceways and wiring.
7. “Coordinate” is defined as to make all arrangements directly with agencies and individuals, confirm schedules, be in attendance at the time work is being carried out, and take full responsibility for having the work carried out correctly and in a timely manner to meet the construction schedule.
8. “Concealed” is defined as hidden from normal sight in ceiling spaces, shafts, walls, partitions, and other furred spaces.
9. “Exposed” is defined as work that is normally visible including areas such as service rooms, mechanical rooms, equipment rooms, and similar spaces.

1.6 Submittals

1. Submittals: in accordance with Section 01 33 00 – Submittal.
2. Submit for review single line electrical diagrams in glazed frames and locate in Electrical Room.
 1. Electrical distribution system in main electrical room.
3. Submit for review fire alarm riser diagram, plan and zoning of building in glazed frames and locate at fire alarm control panel and annunciator.
4. Shop drawings:
 1. Submit drawings stamped and signed by professional engineer registered, or licensed, in the Province of BC, Canada.
 2. Submit wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, wiring conduit, and other items that must be shown to ensure coordinated installation.
 3. Identify on wiring diagrams circuit terminals and indicate internal wiring for each item of equipment and interconnection between each item of equipment.
 4. Indicate of drawings clearances for operation, maintenance, and replacement of operating equipment devices.

5. Submit number of copies indicated in Submittals Section 01 33 00 of drawings and product data to the authority having jurisdiction.
6. If changes are required, notify Engineer of these changes before they are made.
7. Submit a detailed schedule of all shop drawings prior to the first progress draw. Schedule shall include specification section, equipment name, manufacturer's name, and delivery date.
5. Quality Control: in accordance with Section 01 45 00 (01400) – Quality Control
 1. Provide CSA certified equipment and material.
 2. Where CSA certified equipment and materials are not available, submit such equipment and material to inspection authorities for special approval before delivery to site.
 3. Submit test results of installed electrical systems and instrumentation.
 4. Permits and fees: in accordance with General Conditions of contract.
 5. Submit, upon completion of work, load balance report as described in PART 3 – LOAD BALANCE.
 6. Submit certificate of acceptance from authority having jurisdiction upon completion of work to Engineer.
6. Manufacturer's Field Report: submit to Engineer manufacturer's written report, within 3 days of review, verifying compliance of work and electrical system testing, as described in PART 3 – FIELD QUALITY CONTROL.

1.7 Quality Assurance

1. Quality Assurance: in accordance with Section 01 45 00 (1400) – Quality Control.
2. Qualifications: electrical work to be carried out by qualified, licensed electricians who hold valid Master Trade Contractor license in accordance with authorities having jurisdiction.
 1. Employees registered in provincial apprenticeship program: permitted, under direct supervision of qualified licensed electrician, to perform specific tasks.
 2. Permitted activities: determined based on training level attained and demonstration of ability to perform specific duties.
3. Site Meetings:
 1. In accordance with Section 01 32 16 – Construction Schedule.
 2. Site Meetings: schedule site visits, to review work, at stages listed.
 1. After delivery and storage of products, and when preparatory work is complete but before installation begins.
 2. A minimum of twice during progress of work at 50% and 75% complete.
 3. Upon completion of work, after cleaning is carried out.
4. Health and Safety Requirements: do construction occupational health and safety in accordance with Section 01 35 29 (01060) – Health and Safety Requirements.

1.8 Delivery, Storage and Handling

1. Material Delivery Schedule: provide Engineer with schedule of all materials within 2 weeks after award of Contract. Progress claims will not be reviewed until updated schedules are provided.

2. Construction/Demolition Waste Management and Disposal: separate waste materials for reuse and recycling in accordance with Section 01 74 21 – Construction Waste Management.

1.9 System Start-up

1. Instruct Engineer and operating personnel in operation, care and maintenance of systems, system equipment and components.
2. Arrange and pay for services of manufacturer's factory service engineer to supervise start-up installation, check, adjust, balance and calibrate components and instruct operating personnel.
3. Provide these services for such period, and for as many visits as necessary to put equipment in operation, and ensure that operating personnel are conversant with aspects of its care and operation.

1.10 Operating Instructions

1. Provide for each system and principal item of equipment as specified in technical sections for use by operation and maintenance personnel.
2. Operating instructions to include following:
 1. Wiring diagrams, control diagrams, and control sequence for each principal system and item of equipment.
 2. Start up, proper adjustment, operating, lubrication, and shutdown procedures.
 3. Safety precautions.
 4. Procedures to be followed in event of equipment failure.
 5. Other items of instruction as recommended by manufacturer of each system or item of equipment.
3. Print or engrave operating instructions and frame under glass or in approved laminated plastic.
4. Post instructions where directed.
5. For operating instructions exposed to weather, provide weather-resistant materials or weatherproof enclosures.
6. Ensure operating instructions will not fade when exposed to sunlight and are secured to prevent easy removal or peeling.

1.11 Permits and Fees

1. Submit to Electrical Review Department, Local Fire Authorities and Supply Authority the necessary number of drawing and specifications for examination and approval prior to commencement of work. Obtain all required permits and pay all fees.
2. Arrange for review of all work by the authorities having jurisdiction. On completion of the work, furnish final unconditional certificates of approval by the inspecting authorities.

1.12 Equipment Restraint

1. It is the entire responsibility of the equipment manufacturers to design their equipment so that the strength and anchorage of internal components of the equipment exceeds the force level used to restrain and anchor the unit itself to the supporting structure.

1.13 Seismic Analysis and Provisions

1. Install electrical systems with adequate structural support to withstand seismic forces in accordance with Section 4.1.8 of the National Building Code, and Province, Territory or Municipality of the project.

2. Obtain a Structural Engineer licensed in the Province or Territory of the project to perform a review of the proposed electrical installation and prepare installation documents indicating all required seismic supports, bracings and fastenings. These documents shall be sealed and signed by the Structural Engineer and submitted as part of the shop drawing package prior to rough-in work commencing on site.
3. Equipment to be indicated in the structural design documents shall include but not be limited to: suspended transformers, cable trays, suspended conduit runs, free standing distribution equipment such as switchboards, and suspended lighting fixtures.
4. Verify the Facility is classified as “Normal” with an “Importance Factor of 1.0” as referenced in 4.1.8.5 (1) of the National Building Code and Province, Territory or Municipality of the project.
5. Refer to seismic risk reduction of operational and functional components (OFCs) of building S832-06 and meet all requirements.
6. Provide confirmation in writing via a Schedule-C, signed and sealed by the structural engineer, at completion of project that the electrical installation is in general compliance with the structural installation drawings submitted with the shop drawing package.
7. All light fixtures shall be provided with independent chain supports that are fastened to the structure of the Building.
8. The Trade Contractor shall be solely responsible for the full scope of this work. Include all costs of structural design, materials and site review in Bid Price.

1.14 Drawing and Measurements

1. Drawings are generally diagrammatic and are intended to indicate the scope and general arrangement of work and are not detailed installation drawings. Do not scale the drawings. Obtain accurate dimensions from the Architectural and Structural drawings.
2. Consult the Architectural drawings and details for exact locations of fixtures and equipment. Obtain this information from the Engineer where definite locations are not indicated.
3. Take field measurements where equipment and material dimensions are dependent upon building dimensions.

1.15 Project Coordination

1. Check drawings of all trades to verify space and headroom limitations for work to be installed. Coordinate work with all trades and make changes to facilitate a satisfactory installation. Make no deviations from the design intent involving extra cost to the Owner without the Engineer's written approval.
2. The drawings indicate the general location and route to be followed by the electrical services. Where details are not shown on the drawing or only shown diagrammatically, the services shall be installed in such a way as to conserve headroom and interfere as little as possible with the free use of space through which they pass. Service lines shall run parallel to building lines. All services in the ceiling shall be kept as tight as possible to beams or other limiting members at high level. All electrical services shall be coordinated in elevation to ensure that they are concealed in the ceiling or structural space provided unless detailed otherwise on drawings.
3. Work out jointly all interference problems on the site and coordinate all work before fabricating or installing any material or equipment. Where necessary, produce interference/coordination drawings showing exact locations of electrical systems or equipment within service areas, shafts and the ceiling space. Distribute copies of the final interference/coordination drawings to the Engineer and all affected parties.

4. Ensure that all materials and equipment fit into the allotted spaces and that all equipment can be properly serviced and replaced, if and when required. Advise the Engineer of space issues before installing any material or equipment. Demonstrate to the Engineer on completion of the work that all equipment installed can be properly, safely serviced and replaced, if and when required.

1.16 Provision for Future Equipment and Construction

1. Leave clear spaces designated for future equipment or building expansion where indicated. Plan for the installation under this contract and ensure clear, accessible, unhindered access to the space is allowed for.
2. Where contract documents do not clearly indicate the future expansion requirements but known services are required, provide written “request for information” to the Engineer before making assumptions as to intent.

1.17 Sprinkler Proof Requirements

1. All equipment and wiring systems shall be sprinkler proof standard where sprinkler fire protection systems are installed.
2. In rooms where electrical equipment is installed surface-mounted, electrical equipment contained in these rooms is to be protected by non-combustible drip hoods, shields and gasketed doors as applicable to inhibit water ingress into electrical equipment. Exposed conduits connected equipment to utilize water-tight connectors. Top entry to be avoided where possible.
3. In particular, all unit substations, transformers, switchgear, motor control and panelboard shop drawings shall be certified “sprinkler proof” design.

1.18 Phase Construction

1. See specifications and drawings for construction phasing. Make all allowance to phase the work in accordance with project phasing.
2. All existing services and existing building(s) must be maintained in operation. Provide and install temporary services as required.
3. All trades in this division shall make allowance for the implication of having to totally complete all work in the new addition before proceeding with work in the existing building.

1.19 Tender Inquiries

1. All Trade Contractor queries during the tender period shall be made in writing to the Engineer. Trade Contractor queries will be collected and suitable addenda will be issued for clarification. No verbal information will be considered valid or issued by the Engineer's office during tender. All tender queries may be emailed, mailed or couriered to the Engineer's office. No telephone queries will be answered.

1.20 Examination

1. Visit the site before preparing the tender and examine all existing conditions. No extra cost will be considered for any misunderstanding of the work to be done resulting from failure to visit the site.
2. Examine the documents for details of work included. Obtain a written clarification in the event of conflict within the specification, between the specification and the drawing, or in the drawing. Obtain written clarification from the Engineer if work affecting the installation is not clear. Where this is not done in advance, allow in the tender sum for providing the more costly alternative.

1.21 Responsibilities

1. Ensure that equipment does not transmit noise and/or vibration to other parts of the building, as a result of poor installation practices.

2. Where the Contract Documents do not contain sufficient information for the proper selection of equipment for bidding, notify the Engineer during the tendering period. If clarification is not obtainable, allow for the most expensive arrangement. Failure to do this shall not relieve the Trade Contractor of responsibility to provide the intended equipment.
3. Protect equipment and material from the weather, moisture, dust and physical damage.
4. Cover equipment openings and open ends of conduits, piping and pull boxes as work progresses. Failure to do so will result in the Trade Contractor being required to adequately clean or replace materials and equipment at no extra cost to the Owner.
5. Refinish damaged or marred factory finish to factory finish.
6. The specifications and drawings form an integral part of the Contract Documents. Neither the drawings nor the specifications shall be used alone. Work omitted from the drawings but mentioned or reasonably implied in the specifications, and vice versa, shall be considered as properly and sufficiently specified and shall be provided. Misinterpretation of any requirement of either plans or specifications shall not relieve the Trade Contractor of the responsibility of properly completing his trade to the approval of the Engineer.

1.22 Standard of Acceptance

1. Standard of Acceptance means that the item named and specified by the manufacturer and/or catalogue number forms part of the specification and sets standard regarding performance, quality of material and workmanship and when used in conjunction with a reference standard, shall be deemed to supplement the standard.
2. Where two or more manufacturers are listed, the manufacturer's name shown first or underlined or show with a model name and/or number was used in preparing in the base design. Tenders may be based on any one of those named, provided that they meet every aspect of the base design and every aspect of the drawings and specifications.
3. Where other than the first named or the underlined manufacturer or scheduled/specified manufacturer is selected or approved, include the cost of any resulting work (both under this Division and other Divisions) and any necessary redesign or installation or structure. Submit redesign drawing for review with Shop Drawings. Maintain installation, access and servicing clearances. Equipment/materials shall not exceed the available space limitations. Redesign drawings shall be to scale and of a standard equal to the Project Drawings.
4. A visible manufacturer's nameplate shall indicate manufacturer's name, model number, serial number, capacity data, electrical characteristics and approval stamps.

1.23 Progress Claim and Change Order Breakdowns

1. Ten (10) days after the award of contract, submit price breakdowns.
2. In particular cases, more detail may be necessary to properly assess a change order or progress claims. This additional information could include all suppliers and all trade contractors when requested by the Engineer. Provide details for each section of the electrical work listed for each separate electrical change order.
3. Mark-up information is required for change orders but is optional on the original tender price. Change order mark shall be 10% for overhead and 5% for profit.
4. Quotations submitted are to be a detailed and itemized estimate of product, labour, and equipment costs associated with change or revision.

5. Quotations submitted that are not in accordance with requirements above will be rejected and returned for re-submittal. Failure to submit a quotation as per the above requirements will not be grounds for any additional change to contract time.
6. Do not execute any change or revision until written authorization for the change or revision has been obtained from the Consultant.
7. Progress claims will not be certified nor payment made beyond 90% of the overall Electrical contract until commissioning and verification of the systems are complete. This procedure is to allow for any necessary deficiency holdbacks on items which do not become apparent until the systems are commissioned.

1.24 Warranty

1. Use of installed equipment during construction shall not shorten or alter the warranty period, as specified in Division 1.
2. Take note of any extended warranties specified.
3. Furnish a written warranty stating that all work executed under this Division will be free from defects of material and workmanship for a period of one (1) year from the date of substantial performance.
4. Promptly investigate any electrical or control malfunction and repair or replace all such defective work and all other damages thereby which becomes defective during the time of warranty.

1.25 Substantial Performance Requirements.

1. Refer to each section in specifications for detailed requirements.
2. Before the Engineer is requested to make a review for substantial performance of the work:
 1. Commission all systems and prove out all components, interlocks and safety devices.
 2. Submit a letter certifying that all work is complete for the intended use, operational, clean and all required submissions have been completed.
 3. A complete list of incomplete or deficient items shall be provided. If, in the opinion of the Engineer, this list indicates the project is excessively incomplete, a substantial completion review will not be performed.
3. The work will not be considered to be ready for use or substantially complete until the following requirements have been met:
 1. All reported deficiencies have been corrected.
 2. Operating and Maintenance Manuals completed.
 3. Record Drawing markups ready for review.
 4. Systems Commissioning has been completed and has been verified by the Engineer.
 5. All demonstrations to the Owner have been completed.
4. Engineer's Letters of Assurance will not be issued until the following requirements have been met:
 1. All items listed in .1 above have been completed or addressed.
 2. Certificate of Penetrations through Separations.
 3. Provincial or City Electrical Review – Certificate of Review.
 4. Seismic Engineer's Letter of Assurance and Final Review Report.
 5. Certificate of Substantial Performance.

6. Signed off copy of Engineer's Final Review Report.

1.26 Deficiency Holdback and Deficiency Reviews

1. Work under this Division which is still outstanding when substantial performance is certified will be considered deficient and a sum equal to at least twice the estimated cost of completing that work will be held back.
2. It is expected that outstanding work will be completed in an expeditious manner and the entire holdback sum will be retained until the requirements for Total Performance of work have been met and verified.

2. PRODUCTS

2.1 Sustainable Requirements

1. Materials and products to be disposed of responsibly and in accordance with local bylaws.

2.2 Materials and Equipment

1. Provide material and equipment in accordance with Section 01 61 00 Common Product Requirements.
2. Materials and Equipment to be CSA certified. Where CSA certified material and equipment are not available, obtain special approval from inspection authorities before delivery to site and submit such approval as described in PART 1 – SUBMITTALS.
3. Factory-assemble control panels and component assemblies.

2.3 Warning Signs

1. Warning Signs: in accordance with requirements of Authority Having Jurisdiction.
2. Porcelain enamel signs, minimum size 175 x 250 mm.
3. For high voltage installations, provide permanent, legible warning notice carrying the wording: "DANGER – HIGH VOLTAGE" as required and indicated in Section 36 of CSA C22.1, CEC – Part 1.

2.4 Wiring Terminations

1. Ensure lugs, terminals, screws used for termination of wiring are suitable for either copper or aluminum conductors.

2.5 Equipment Identification

1. Identify electrical equipment with nameplates and labels as follows:
 1. Nameplates: lamacoid 3mm thick plastic engraving sheet, matte white finish face, black core, lettering accurately aligned and engraved into core and mechanically attached with self-tapping screws.
2. Size as follows:

NAMEPLATE SIZES			
Size 1	10 x 50 mm	1 line	3 mm high letters
Size 2	12 x 70 mm	1 line	5 mm high letters
Size 3	12 x 70 mm	2 lines	3 mm high letters
Size 4	20 x 90 mm	1 line	8 mm high letters
Size 5	20 x 90 mm	2 lines	5 mm high letters
Size 6	25 x 100 mm	1 line	12 mm high letters

Size 7	25 x 100 mm	2 lines	6 mm high letters
--------	-------------	---------	-------------------

2. Labels: embossed plastic labels with 6 mm high letters unless specified otherwise.
3. Wording on nameplates and labels to be approved by Engineer prior to manufacture.
4. Allow for minimum of twenty-five (25) letters per nameplate and label.
5. Nameplates for terminal cabinets and junction boxes to indicate system and/or voltage characteristics.
6. Disconnects, starters and contactors: indicate equipment being controlled and voltage.
7. Terminal cabinets and pull boxes: indicate system and voltage.
8. Transformers: indicate name of transformer name, capacity, primary and secondary voltages.

2.6 Wiring Indication

1. Identify wiring with permanent indelible identifying markings, numbered and coloured plastic tapes, on both ends of phase conductors of feeders and branch circuit wiring.
2. Maintain phase sequence and colour coding throughout.
3. Colour coding: to CSA C22.1 (current edition).
4. Use colour coded wires in communication cables, matched throughout system.

2.7 Conduit and Cable Identification

1. Colour code conduits, boxes and metallic sheathed cables.
2. Code with plastic tape or paint at points where conduit or cable enters wall, ceiling, or floor, and at 15 m intervals.
3. Colour: 25 mm wide prime colour and 20 mm auxiliary.

	Prime	Auxiliary
120/208 V Normal System	Blue	
120/208 V Emergency System	Blue	Red
347/600 V Normal System	Orange	
347/600 V Emergency System	Orange	Red
Up to 25 kV	Black	
Communication Systems	Green	
Fire Alarm	Red	
Security Systems	Red	Brown

2.8 Finishes

1. Shop finish metal enclosure surfaces by application of rust resistant primer inside and outside, and at least two coats of finish enamel.
 1. Indoor switchgear and panelboards in service rooms shall be painted with ASA 61 Grey finish.

2.9 Fire-Stopping

1. Include Labor, materials and equipment necessary to complete the installation required for Firestopping of Through Penetrations in Fire Rated Assemblies.

2. Install as per manufacturer's instructions and only use assemblies that are suitable for the specific application.
3. Related Sections
 1. Consult all other Sections, determine the extent and character of related work and properly coordinate work specified herein with that specified elsewhere to produce a complete installation.
 2. This shall be considered as an augmentation to Section 07 84 00 or any of its sub-sections of Section 07 84 00.
 3. This addresses those unique elements that affect the Firestopping of Information Technology cabling systems which may not be addressed in Section 07 84 00 or any of its sub-sections.
4. Reference
 1. CAN4-S115-M, "Standard Method of Fire Tests of Firestop Systems".
 2. CAN/ULC-S115, "Standard Method of Fire Tests of Firestop Systems".
 3. Underwriters Laboratories Canada Inc. (ULC) – Fire Resistance Directory
 4. Underwriters Laboratories, Inc. (CUL) – Fire Resistance Directory of Products Certified for use in Canada.
5. Performance
 1. Fire rated cable pathway devices shall be used for ALL low-voltage, video, data and voice cabling, optical fibre raceways and certain high-voltage cabling where frequent cable moves, adds and changes may occur. Pathways required for high-voltage cabling will be detailed on the prints. Such devices shall:
 1. Meet the hourly fire-rating of fire rated wall and or floor penetrated.
 2. Be tested for the surrounding construction and cable types involved.
 3. Have ULC, cUL or cULus Systems permitting cable loads from; "Zero to 100% Visual Fill." This requirement eliminates need for fill-ratio calculations to be made by cable technicians to ensure cable load is within maximum allowed by ULC, cUL or cULus System.
 4. Not have a constrictive inner liner that constricts around or compresses cables tightly together encouraging potential alien cross-talk or network interference.
 5. Be "Zero-Maintenance", zero-maintenance is defined as; No action required by cabling technician to open and/or close pathway for cable moves, adds or changes, such as, but not limited to:
 1. Opening or closing of doors.
 2. Spinning rings to open or close fabric liner.
 3. Removal and or replacement of any material such as, but not limited to, firestop caulk, putty, pillows, bags, foam muffins, foam foam plugs, foam blocks, or foam closure of any sort.
 4. Furnish letter from manufacturer certifying compliance with this definition of "Zero-Maintenance."
 6. Pathways shall be engineered such that two or more devices may be ganged together. When installed in gypsum wallboard walls, pathways shall be ganged and supported by a grid designed to transfer the weight load of the pathways to the studs by anchoring with fasteners.

Unless tested otherwise by an accredited third-party test lab such as ULC, multiple pathways SHALL NOT be clustered unless separated by a minimum of 6" on all sides.

7. Pathways shall be engineered to be re-enterable permitting retrofit and removal from around existing cables without cutting and re-splicing them.
8. Cable Pathway Devices passing vertically through floors shall have equal FT Rating. (See cUL System # F-A-3037, Item #4 "EZ-PATH Grid T-Rating Kit" Part # TRK444)
9. Affix adhesive wall label immediately adjacent to devices to communicate to future cable technicians, authorities having jurisdiction and others the manufacturer of the device and the corresponding cUL, ULC or cULus System number installed.
2. Where non-mechanical pathways cannot be utilized, such as sealing (caulking) around single or grouped conduits, provide products that upon curing do no re-emulsify, dissolve, leach, breakdown or otherwise deteriorate over time from exposure to atmospheric moisture, sweating pipes, ponding water or other forms of moisture characteristic during or after construction. Provide letter from manufacturer certifying compliance with this section.
3. Cable Pathway shall replace conduit sleeves in walls and floors, and;
 1. When installed individually in floors, devices shall pass through core-drilled opening utilizing tested floor plates.
 2. When multiple units are ganged in floors, devices shall be anchored by means of a tested grid.
 3. When installed individually in walls, devices shall pass through core drilled opening utilizing tested wall plates.
 4. When multiple units are ganged in walls, devices shall be anchored by means of a tested grid.
4. Cable tray shall terminate each fire barrier and resume on the other side such that cables pass independently through devices. Cable tray shall be properly supported on each side of fire barrier.
6. Submittals
 1. Submit under provisions of Section 01 30 00.
 2. Product Data: Provide manufacturer's standard catalog data for specified products demonstrating compliance with referenced standards and listing numbers of systems in which each product is to be used.
 3. Schedule of ULc, cUL or cULus System Drawings: Submit schedule of all expected opening locations and sizes, penetrating items, and required listed design numbers to seal openings to maintain fire resistance ratings. If engineering recommendations are necessary, list these in the schedule too.
 4. ULC, cUL or cULus System Drawings: Furnish copies of all ULC, cUL or cULus Systems identified in schedule above. Include any engineering recommendations.
 5. Certificates: Product Certificate of Compliance from the firestop system manufacturer certifying material compliance with applicable code and specified performance characteristics.
 6. Installation Instructions: Submit Manufacturer's printed installation instructions.
7. Quality Assurance
 1. Products/Systems: Provide firestopping systems that comply with the following requirements:

1. Firestopping tests are performed by a qualified, testing and inspection agency. A qualified testing and inspection agency is ULC, cUL or cULus, or another agency performing testing and follow-up inspection services for firestop system acceptable to authorities having jurisdiction.
2. Firestopping products bear the classification marking of qualified testing and inspection agency.
2. Installer Qualifications: Experience in performing work of this section who is qualified by the firestopping manufacturer as having been provided the necessary training to install firestop products in accordance with specified requirements.
3. Field Inspections: Inspection of completed work shall be performed by the Consultant and Authority having jurisdiction (AHJ), and/or the building underwriter's designee. If required by AHJ or underwriter, inspections may be performed by an independent, third-party construction inspection and testing service provided that:
 1. Inspections are performed to the requirements of the following standards, as applicable.
 1. Construction Joints: ASTM E2393 – 10a Standard Practice for On-Site Inspection of Installed Fire Resistive Joint Systems and Perimeter Fire Barriers.
 2. Service Penetrations: ASTM E2174 – 10a Standard Practice for On-Site Inspection of Installed Fire Stops.
 2. Individual(s) performing inspection SHALL provide evidence of valid Errors and Omissions insurance coverage for this service.
 3. Individual(s) performing inspection SHALL NOT have any financial connection to installer, especially including firestop manufacturer, distributor or supplier.
8. Project Conditions
 1. Do not install firestopping products when ambient or substrate temperatures are outside limitations recommended by manufacturer.
 2. Do not install firestopping products when substrates are wet due to rain, frost, condensation, or other causes.
 3. Maintain minimum temperature before, during, and for a minimum 3 days after installation of materials.
 4. Do not use materials that contain flammable solvents.
 5. Coordinate construction of openings and penetrating items to ensure that through-penetration firestop systems are installed according to specified requirements.
 6. Coordinate sizing of sleeves, openings, core-drilled holes, or cut openings to accommodate through-penetration firestop systems.
 7. Schedule installation of firestopping after completion of penetrating item installation but prior to covering or concealing of openings.
9. Acceptable Manufacturers
 1. Single Source: Obtain firestop systems for each type of penetration and construction condition indicated only from a single manufacturer.
 2. Acceptable Manufacturer: Hilti, 3M, Tremco, PFP or approved equal.
10. Materials

1. General: use only firestopping products that have been tested for specific fire resistance rated construction conditions conforming to construction assembly type, penetrating item type, annular space requirements, and fire rating involving for each separate instance.
2. Firestop Sealants: Single component latex formulations that upon cure do not re-emulsify during exposure to moisture.
3. Firestop Putty: Intumescent, non-hardening, water resistant putties containing no solvents, inorganic fibers or silicone compounds.
4. Firestop Pillows: Re-enterable, non-curing, mineral fiber core encapsulated on six sides with intumescent coating contained in flame retardant poly bag.
5. Fire-Rated Cable Grommet: Firestop Grommet is a molded, two-piece grommet with an integral fire and smoke sealing foam membrane for sealing individual cables or pairs of cables penetrating through framed wall assemblies. Grommet snaps together around cable and locks tightly into the wall.

11. Examination

1. Before beginning installation, verify that substrate conditions previously installed under other sections are acceptable for installation of firestopping in accordance with manufacturer's installation instructions and technical information.
2. Surfaces shall be free of dirt, grease, oil, scale, laitance, rust, release agents, water repellants, and any other substances that may inhibit optimum adhesion.
3. Provide masking and temporary covering to protect adjacent surfaces.
4. Do not proceed until unsatisfactory conditions have been corrected.

12. Installation

1. General: Install through-penetration firestop systems in accordance with Performance Criteria and in accordance with the conditions of testing and classification as specified in the published design.
2. Manufacturer's Instructions: Comply with manufacturer's instructions for installation of firestopping products.

13. Field Quality Control

1. Inspections: Owner shall engage qualified independent inspection agency to inspect through-penetration firestop systems.
2. Keep areas of work accessible until inspection by Authorities Having Jurisdiction.
3. Where deficiencies are found, repair firestopping products so they comply with requirements.

14. Documentation

1. Place system label on each side of wall penetrations.
2. Place a reproduction (photocopy) of the ULC System description in a document protector and mount to the wall next to the wall penetration.
 1. Highlight the section of the system description that lists the allowed cable types.

2.10 Bases Supports

1. Where conduit and equipment is located on walls or slabs which will not permit the support of equipment, provide suitable supports to the building structure. Supports shall be constructed of steel members or of steel pipe and fittings designed to safely support the equipment.

2. All equipment bases shall be set on pads of kinetic pre-compressed fiberglass or vibration isolators sized to suit the equipment which they ought to support.

2.11 Inserts, Sleeves and Curbs

1. Provide all inserts, sleeves and curbs required for the work of this contract.
2. Use only factory-made threaded or toggle type inserts as required for support and anchors, properly sized for the load to be carried, Place inserts only in portions of the main structure and not in any finishing material.
3. Use factory-made expansion shields where inserts cannot be placed, but only where approved by the Engineer and only for loads of 50 kg or less.
4. Do not use power-activated tools unless with written permission of the Engineer.
5. Supply and locate all inserts, holes, anchor bolts and sleeves in time when walls, floors and roof are erected.
6. Size sleeves to provide 25 mm clearance all around.
7. Use the following sleeving materials:
 1. Through all interior walls, use Schedule 40 steel pipes, machine cut, flush with finished structure. Check room finish schedules.
 2. Through all exterior walls above grade, use Schedule 40 steel pipes, machine cut, flush with finished structure inside and to suite flashing on outside.
 3. Through all exterior walls below grade and all other waterproof walls, use wrought iron pipes. Check flashing details for further information.
 4. Through all waterproof floors, through washrooms, janitor's closets, boiler rooms, mechanical rooms, kitchen and through roofs, use wrought iron sleeves, machine cut. Extend sleeves 100 mm above finished floor upward and cut flush with underside of floor.
 5. Approved type plastic sleeves, conduit sleeves or 18-gauge galvanized steel sleeves may be used as an alternative for Schedule 40 steel sleeves in interior areas.
 6. Provide 100 mm high, 100 mm wide water-tight concrete curbs with 20 mm chamfered edges around all sleeves passing through waterproof floors except where furred in.
 7. Sleeves are not required in walls and dry area floors where conduit is installed ahead of wall construction.
 8. Pack all sleeves between the conduit or cable passing through the sleeve and the sleeve and all spare sleeves with loose fiberglass insulation. Seal and annular space on both sides as follows:
 1. For all horizontal sleeves in exposed areas, use a seal of equal or better fire rating than the wall to be sealed.
 2. For all horizontal concealed sleeves through firewalls and through walls separating areas of different air pressure, use a permanently resilient silicone base or equal sealing compound.
 3. For all vertical sleeves through roofs, washrooms, janitor closets, equipment rooms, use permanently resilient silicone base or equal compound, non-flammable and waterproof. Ensure that the seal is compatible with floor and ceiling finishes. Check the room finishes schedules for further information.
 4. The Trade Contractor to provide sleeving diagrams/drawings to the structural Engineer for review and approval prior to any work commencing.

2.12 Cutting and Patching

1. The Trade Contractor shall employ the particular trades to do all required cutting and patching and the repairing of surfaces for his work.
2. Supporting members of any floor, wall or the building structure shall be cut only in such a location and manner as directed by the Engineer or the Structural Engineer.
3. Provide fire barriers around all components in holes which penetrate fire separations. The fire barrier medium provided shall make the fire separation equal to or better than the one which was cut away. All materials shall be CSA approved and UL listed.
4. All floor saw cutting and drilling required for electrical services is to be performed within hours approved by Owner. Prior written notice of 48 hours must be given to Owner.

2.13 Access Panels and Doors

1. Install concealed electrical equipment requiring adjustment or maintenance in locations easily accessible through access panels or doors. Install systems and components to result in a minimum number of access panels. Indicate access panels on "Record" drawings.
2. Provide other Trades with panels, doors, or the frame therefore, complete with all pertinent information for installation. Supply to Trade in whose work they occur to install them. Ensure that access doors are installed in a manner to match the building material grids where applicable.
3. Prepare detail drawings showing location and type of access doors in coordination with other trades before proceeding with installation and hand these to the Engineer for review.
4. Size access doors to provide adequate access and commensurate with the type of structure and architectural finish. Should it be necessary for persons to enter, provide minimum 610 mm x 610 mm size doors.
5. Ensure proper fire rating of access doors in fire separations.
6. Access doors shall be Maxam Metal or Acudor with 14 US gauge steel door panel, rust-resistant concealed hinges and positive locking with self-opening screwdriver-operated lock. Frame shall be suitable for wall installation and shall have integral keys for plaster walls. Doors in the tile wall shall be stainless steel and ceiling shall be suitable for plaster covering with only the frame joint showing. All other doors shall be prime painted steel. Minimum size of doors shall be 12" x 18" (300 mm x 450 mm). Wherever possible, 24" x 24" (600 mm x 600 mm) doors shall be used.
7. Lay-in type tiles, properly marked, may serve as access panels. Marking of ceiling tiles must be coordinated with Owner.
8. Panels in glazed tile walls shall be 12-gauge, 304 alloy stainless steel, No. 4 finish, with recessed frame secured with stainless steel counter-sunk flush head screws.
9. Panels in plaster surfaces shall have dish-shaped door and welded metal lath, ready to take plaster. Provide a plastic grommet for door key access.
10. Other access doors shall be welded 12-gauge steel, flush type with concealed hinges, lock and anchor straps, complete with factory prime coat. Obtain approval of non-standard door construction details.

2.14 Backboards

1. Provide plywood backboard for all electrical rooms, communication rooms and security rooms as indicated. Where no size is indicated, provide a backboard a minimum of 100 mm wider and 100 mm higher than the equipment. Where more than one piece of equipment is installed on the backboard,

construct the backboard of a size to suit the maximum vertical and horizontal dimensions of the equipment.

2. For communication and security rooms where not indicated, provide backboard for all surfaces in communication and security rooms. Backboard shall extend from finished floor to 8' AFF, continuously around the communication and security rooms, or as indicated.
3. Construct plywood backboards from 20 mm thick fir plywood (FSC to UF Free) good one side.
4. Use fire-retardant backboards, pressure impregnated with fire-retardant chemicals, and stamp. Conform to CSA 080.
5. Fastenings:
 1. Fasten each backboard to a wall or to a support structure using cadmium-plated hardware. Provide a flat washer under the head of each fastener. Recess the head of the mounting bolt where equipment, including future equipment, is to be installed.
 2. Use expansion shields, toggle bolts or other types of wall fastenings to suit the wall type. Align the mounting bolts with the wall studs for stud type walls.
 3. Install fastenings to a maximum 500 mm apart in both the vertical and horizontal directions.
 4. When installing equipment heavier than 50 kg, fasten the equipment through the backboard directly to the wall or support structure.

3. EXECUTION

3.1 Installation

1. Do complete installation in accordance with CSA C22.1 except where specified otherwise.
2. Do overhead and underground systems in accordance with CSA C22.3 No. 1 except where specified otherwise.

3.2 Nameplates and Labels

1. Ensure manufacturer's nameplates, CSA labels and identification nameplates are visible and legible after equipment is installed.

3.3 Locations of Devices

1. Locate outlets in accordance with Section 26 05 32 – Outlet Boxes, Conduit Boxes and Fittings.
2. Do not install outlets back-to-back in wall; allow minimum 150 mm horizontal clearance between boxes.
3. Change location of devices at no extra cost or credit, provided distance does not exceed 3000 mm, and information is given before installation.
4. Locate light switches on latch side of doors.
5. Locate disconnect devices in mechanical and elevator machine rooms on latch side of floor.

3.4 Mounting Heights

1. Mounting height of equipment is from finished floor to centerline of equipment unless specified or indicated otherwise.
2. If mounting height of equipment is not specified or indicated. Verify before proceeding with installation.
3. Install electrical equipment at following heights unless indicated otherwise.

1. Local switches: 1400 mm.
2. Wall receptacles:
 1. General: 300 mm.
 2. Above top of continuous baseboard heater: 200 mm.
 3. Above top of counters or counter splash backs: 175 mm.
 4. In mechanical rooms: 1400 mm.
3. Panelboards: 1800 mm to top or as required by Code or as indicated.
4. Communication outlets: 300 mm.
5. Wall mounted telephone and interphone outlets: 1500 mm
6. Doorbell pushbuttons: 1500 mm
1. Fire alarm manual stations: not less than 1,050 mm and not more than 1,150 mm to centre of device.
7. Fire alarm bells: not less than 2300 mm and shall maintain 150 mm clearance from the ceiling.
8. Fire Visible Signal Device: not less than 2000 mm and not more than 2400 mm of the entire lens.
9. Fire fighter handset: not less than 1350 mm and not more than 1500 mm to centre of device.
10. Fire alarm end of line resistors: less than 1800 mm to centre of device.
11. Annunciator: not more than 1800 to top of display.
12. Wall-mounted emergency lighting heads: 2.35 m.

3.5 Coordination of Protective Devices

1. Ensure circuit protective devices such as overcurrent trips, relays and fuses are installed to required values and settings.

3.6 Field Quality Control

1. Load Balance:
 1. Measure phase current to panelboards with normal loads (lighting) operating at time of acceptance; adjust branch circuit connections as required to obtain best balance of current between phases and record changes.
 2. Measure phase voltages at loads and adjust transformer taps to within 2% of rated voltage of equipment.
 3. Provide upon completion of work, load balance report as directed in PART 1 – SUBMITTALS: phase and neutral currents on panelboards, dry-core transformers and motor control centres, operating under normal load, as well as hour and date on which each load was measured, and voltage at time of test.
2. Conduct following Tests:
 1. Power generation and distribution system including phasing, voltage, grounding and load balancing.
 2. Circuits originating from branch distribution panels.
 3. Motors, heaters and associated control equipment including sequenced operation of systems where applicable.

4. Insulation resistance testing:
 1. Megger circuits, feeders and equipment up to 350 V with a 500 V instrument.
 2. Megger 350-600 V circuits, feeders and equipment with a 1000 V instrument.
 3. Check resistance to ground before energizing.
 3. Carry out tests in presence of Engineer.
 4. Provide instruments, meters, equipment and personnel required to conduct tests during and at conclusion of project.
 5. Manufacturer's Field Services:
 1. Obtain written report from manufacturer verifying compliance of work, in handling, installing, applying, protecting and cleaning of product and submit Manufacturer's Field Reports as described in PART 1 – SUBMITTALS.
 2. Provide manufacturer's field services consisting of product use recommendations and periodic site visits for review of product installation in accordance with manufacturer's instructions.
 3. Schedule site visits, to review work, as directed in PART 1 – QUALITY ASSURANCE.
 6. Reports:
 1. Provide written reports in a timely manner upon completion of testing and load balance. Indicate date and hour tested.
- 3.7 Cleaning
1. Clean and touch up surfaces of shop-painted equipment scratched or marred during shipment or installation, to match original paint.
 2. Clean and prime exposed non-galvanized hangers, racks and fastenings to prevent rusting.

END OF SECTION 26 05 00

GENERAL

1.1 Section Includes

1. Materials and Installation for wire and box connectors.

1.2 Related Sections

1. This section of the specification forms part of the contract documents and is to be read, interpreted and coordinated with all other parts.
2. Section 01 74 21 – Construction/Demolition Waste Management and Disposal.

1.3 References

1. Canadian Standards Association (CSA International):
 1. CAN/CSA-C22.1 (current edition), Canadian Electrical Code, Part 1.
 2. CAN/CSA-22.2 No. 18 (current edition), Outlet Boxes, Conduit Boxes, Fittings and Associated Hardware.
 3. CSA C22.2 No. 65 (current edition), Wire Connectors.
2. Electrical and Electronic Manufacturer's Association of Canada (EEMAC):
 1. EEMAC 1Y-2, 1961 Bushing Stud Connectors and Aluminum Adapters (1,200 Ampere Maximum Rating).
3. National Electrical Manufacturer's Association (NEMA).

1.4 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.5 Submittals

1. Submittals: in accordance with Section 01 33 00 – Submittal Procedures.
2. Product Data:
 1. Provide manufacturer's printed product literature, specifications and data sheets, and include product characteristics, performance criteria, physical size, finish and limitations.

1.6 Delivery, Storage and Handling

1. Deliver, store, handle and protect materials in accordance with Section 01 61 00 – Common Product Requirements.
2. Deliver, store and handle materials in accordance with manufacturer's written instructions.
3. Handle materials with suitable lifting equipment.
4. Store materials in heated, dry, weather-protected enclosure.

2. PRODUCT

2.1 Material

1. Pressure type wire connectors shall be in accordance with CSA C22.2 No. 65, with current carrying parts of copper and aluminum sized to fit copper and aluminum conductors as required.
2. Fixture type splicing connectors shall be in accordance with CSA C22.2 No. 65, with current carrying parts of copper sized to fit copper conductors 10 AWG or less.
3. Bushing Stud Connectors shall be NEMA to consist of:
 1. Connector body and stud clamp for stranded copper and aluminum conductors.
 2. Clamp for stranded copper conductors.
 3. Clamp for stranded aluminum conductors.
 4. Stud clamp bolts.
 5. Bolts for copper conductors.
 6. Bolts for aluminum conductors.
 7. Sized for conductors as indicated.
4. Clamps or connectors for armoured cable, aluminum sheathed cable, flexible conduit, and non-metallic sheathed cable as required shall be in accordance with CAN/CSA-C22.2 No. 18.

2. EXECUTION

2.1 Installation

1. Remove insulation carefully from ends of conductors and:
 1. Apply coat of zinc joint compound on aluminum conductors prior to installation of connectors.
 2. Install mechanical pressure type connectors and tighten screws with appropriate compression tool recommended by manufacturer. Installation shall meet secureness tests in accordance with CSA C22.2 No.65.
 3. Install fixture type connectors and tighten. Replace insulating cap.
 4. Install bushing stud connectors in accordance with NEMA.

2.2 Maintenance – Clearances

1. Provided clearance around systems, equipment and components for observation of operation, review, servicing, maintenance and as recommended by manufacturer and CEC, Part 1.

2.3 Cleaning

1. Proceed in accordance with Section 01 74 11 – Cleaning.
2. On completion and verification of performance of installation, remove surplus materials, excess materials, rubbish, tools and equipment.

END OF SECTION 26 05 20

1. GENERAL

1.1 Section includes

1. Materials and installation for Wires and Cables (0 – 600 V)

1.2 Related Sections

1. This section of the specification forms part of the contract documents and is to be read, interpreted, and coordinated with all other parts.

1.3 References

1. Canadian Standards Association (CSA International)
 1. CSA C22.1 (current edition), Canadian Electric Code, Part 1.
 2. CSA C22.2 No .0.3-09, Test Methods for Electrical Wires and Cables.
 3. CAN/CSA-C22.2 No. 131-M89 (current edition), Type TECK 90 Cable

1.4 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.5 Submittals

1. Submittals: in accordance with Section 01 33 00 – Submittal Procedures.
2. Product Data:
 1. Provide manufacturer's printed product literature, specifications and data sheets, and include product characteristics, performance criteria, physical size, finish and limitations.
 2. Provide product data in accordance with Section 01 33 00 – Submittal Procedures.

1.6 Delivery, Storage and Handling

1. Deliver, store, handle and protect materials in accordance with Section 01 61 00- Common Product Requirements.
2. Deliver, store and handle materials in accordance with manufacturer's written instruction.
3. Handle materials with suitable lifting equipment.
4. Store materials in heated, dry, weather-protected enclosure.

2. PRODUCTS

2.1 Building Wires

1. All conductors shall be copper unless requested in the contract documents. When indicated only larger than No. 2 AWG can be Aluminum.
2. Conductors shall be stranded for 10 AWG and larger. Minimum size shall be 12 AWG.
3. Copper conductors shall be sized as indicated, with 600V insulation of cross-linked thermosetting polyethylene material rated 75 °C.

4. Copper conductors shall be sized as indicated with thermoplastic insulation type T90 Nylon rated at 600 V.

2.2 Teck 90 Cable

1. Cable shall be in accordance with Section 26 05 00 – Common work results for Electrical.
2. Conductors:
 1. Grounding conductor shall be copper, sized as indicated.
 2. Circuit conductors shall be copper or aluminum as indicated, size as indicated.
3. Insulation:
 1. Ethylene propylene rubber EP.
 2. Cross-linked polyethylene XLPE.
 3. Rating shall be 600 V.
4. Inner jacket shall be polyvinyl chloride material.
5. Armour shall be aluminum or steel interlocked.
6. Outer jacket: Sunlight Resistant AG14 Polyvinyl chloride jacket, -40 C, FT-4 rated suitable for direct burial and suitable for Class 1, Zone 1 and 2, and classes 2, and 3, Division 1 and 2 hazardous locations.
7. Fastenings:
 1. One-hole steel straps to secure surface cables 50 mm and smaller. Two-hole steel straps for cables larger than 50 mm.
 2. Threaded rods shall be 6 mm diameter to support suspended channels.
8. Connectors:
 1. Watertight approved for Teck cable.

2.3 Armoured Cables

1. Conductors shall be insulated, copper, size as indicated.
2. Type shall be AC90 or ACWU90.
3. Armour shall be interlocking type fabricated from aluminum strip.
4. Connectors shall be anti-short connectors.

2.4 Aluminum Sheathed Cable

1. Conductors shall be copper, size as indicated.
2. Insulation shall be cross linked polyethylene Type RA90 rated 600 V.
3. Sheath shall be aluminum applied to form continuous corrugated sheath.
4. Outer jacket shall be thermoplastic applied over sheath and to be compliant to applicable Building Code classification for this project
5. Fastenings for aluminum sheathed cable:
 1. One-hole steel straps to secure surface cables 25 mm and smaller. Two-hole steel straps for cables larger than 25 mm. Use aluminum strap only with single conductor cable.
 2. Threaded rods shall be 6 mm diameter to support suspended channels.

2.5 Controls Cables

1. Type shall be LVT with two soft annealed copper conductors, sized as indicated.
 1. Insulation shall be cross-linked polyethylene XLPE.
 2. Sheath shall be thermoplastic jacket.
2. Type shall be low energy 300 V control cable with solid annealed copper conductors sized as indicated LVT with two soft annealed copper conductors, sized as indicated:
 1. Insulation shall be cross-linked polyethylene XLPE.
 2. Shielding shall be tape coated with paramagnetic material over each pair.
 3. Overall covering shall be PVC jackets.
3. Type shall be 600 V stranded annealed copper conductors, sizes as indicated:
 1. Insulation shall be cross-linked polyethylene XLPE.
 2. Shielding shall be magnetic tape.
 3. Overall covering shall be thermoplastic jacket.

3. EXECUTION

3.1 General Cable Installation

1. Install cable in trenches in accordance with Section 33 71 73.02 – Underground Electrical Service.
2. Terminate cables in accordance with 26 05 20 – Wire and Box Connectors – (0 – 1,000V).
3. Cable Colour Coding shall be in accordance with Section 26 05 00 – Common work Results for Electrical.
4. Conductor length for parallel feeders to be identical.
5. Lace or clip groups of Feeder Cables at distribution centres, pull boxes, and termination points.
6. Wiring in walls shall be typical drop or loop vertically from above to better facilitate future renovations. Generally wiring from below and horizontal wiring in walls to be avoided unless indicated.
7. Branch circuit wiring for surge suppression receptacles and permanently wired computer and electronic equipment to be two-wire circuits only, i.e. common neutrals not permitted.
8. Provide numbered wire collars for control wiring. Numbers to correspond to control shop drawing legend. Obtain wiring diagram for control wiring.

3.2 Installation for Building Wires

1. Install wiring as follows:
 1. In conduit systems in accordance with Section 26 05 34 – conduits, Conduit Fastenings and Conduit Fittings.
 2. In underground ducts in accordance with Section 33.
 3. In underfloor distribution system, in accordance with Section 21.
 4. In wireways and auxiliary gutters in accordance with Section 21.

3.3 Installation of Teck 90 Cable (0 – 1,000 V)

1. Group cables wherever possible on channels.
2. Install cable concealed where possible, securely support by straps.

- 3.4 Installation of Armoured Cables
 - 1. Group cables wherever possible on channels.
- 3.5 Installation of Aluminum Sheathed Cable
 - 1. Group cables wherever possible on channels.
- 3.6 Installation of Control Cables
 - 1. Install control cables in conduit.
 - 2. Ground control cable shield.
- 3.7 Cleaning
 - 1. Proceed in accordance with Section 01 74 11 – Cleaning.
 - 2. On completion and verification of performance of installation, remove surplus materials, excess materials, rubbish, tools and equipment.
- 3.8 Field Quality Control
 - 1. Perform tests in accordance with Section 26 05 00 – Common work Results for Electrical.
 - 2. Perform tests before energizing electrical system.

END OF SECTION 26 05 21.02

1. **GENERAL**

1.1 Section Includes

1. Materials and installation for connectors and terminations.

1.2 Related Sections

1. This section of the specification forms part of the contract documents and is to be read, interpreted and coordinated with all other parts.
2. Section 01 33 00 – Submittal Procedures.
3. Section 01 74 21 – Construction/Demolition Waste Management and Disposal.
4. Section 26 05 33 – Raceway and Boxes for Electrical Systems.

1.3 References

1. Canadian Standards Association (CSA International):
 1. CSA C22.1 (current edition), Canadian Electrical Code, Part 1.
 2. CSA C22.2 No .0.3-09, Test Methods for Electrical Wires and Cables.
 3. CSA C22.2 No. 41 (current edition), Grounding and Bonding Equipment.

1.4 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.5 Submittals

1. Submittals: in accordance with Section 01 33 00 – Sustainable Procedures.
2. Product Data:
 1. Provide manufacturer's printed product literature, specifications and data sheets, and include product characteristics, performance criteria, physical size, finish and limitations.
 2. Submit product data in accordance with Section 01 33 00 – Submittal Procedures.

1.6 Delivery, Storage and Handling

1. Deliver, store, handle and protect materials in accordance with Section 01 61 00 – Common Product Requirements.
2. Deliver, store and handle materials in accordance with manufacturer's written instructions.
3. Handle materials with suitable lifting equipment.
4. Store materials in heated, dry, weather-protected enclosure.

1.7 Certificates

1. Obtain inspection certificate of compliance covering high voltage stress coning from Inspection Authority and include it with maintenance manuals.

2. **PRODUCT**

2.1 Connectors and Terminations

1. Copper short barrel compression connectors to CSA C22.2 No. 65 as required sized of conductors.
2. Contact aid for aluminum cables where applicable.

3. **EXECUTION**

3.1 Installation

1. Install stress cones, terminations, and splices in accordance with manufacturer's instructions.
2. Bond and ground as required to CSA C22.2 No.41.

3.2 Maintenance – Clearances

1. Provide clearance around systems, equipment and components from observation of operation, review, servicing, maintenance and as recommended by manufacturer and CEC, Part 1.

3.3 Cleaning

1. Proceed in accordance with Section 01 74 11 – Cleaning.
2. On completion and verification of performance or installation, remove surplus materials, excess materials, rubbish, tools and equipment.

END OF SECTION 26 05 22

1. **GENERAL**

1.1 Section Includes

1. Materials and Installation for secondary grounding.

1.2 Related Sections

1. This section of the Specification forms part of the contract documents and is to be read, interpreted and coordinated with all other parts.

1.3 References

1. American National Standards Institute (ANSI)/Institute of Electrical and Electronics Engineers (IEEE).
2. ANSI/IEEE 837 (current edition), Qualifying Permanent Connections Used in Substation Grounding.
3. CSA C22.2 No.0.4 (current edition) – Bonding and Ground of Electrical Equipment (Protective Grounding).

1.4 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.5 Delivery, Storage, and Handling

1. Deliver, store, handle and protect materials in accordance with Section 01 61 00 – Common Product Requirements.
2. Deliver, store and handle materials in accordance with manufacturer's written instructions.
3. Handle materials with suitable lifting equipment.
4. Store materials in heated, dry, weather-protected enclosure.

1.6 Testing Requirements

1. Perform ground continuity and resistance test using method appropriate to site conditions. Measure ground grid resistance.
2. Any third part testing agency costs for the testing and reporting shall be included in the Electrical Division base tender and shall be carried out by a pre-approved testing agency.

2. **PRODUCTS**

2.1 Material

1. Clamps for grounding of conductor: size as required to electrically conductive underground water pipe.
2. Copper conductor: minimum 6 m long for each concrete encased electrode, bare, stranded, soft annealed, size as indicated.
3. Rod electrodes: galvanized steel 19 mm diameter by 3 m long.
4. Plate electrodes: galvanized steel, surface area 0.2 m², 1.6 mm thick.
5. Grounding conductors: bare stranded copper as indicated.
6. Insulated grounding conductors: green, type RWU90 XLPE.

7. Ground bus: copper, size as indicated, complete with insulated supports, fastenings, connectors.
8. Non-corroding accessories necessary for grounding system, type, size, material as indicated, including but not necessarily limited to:
 1. Grounding and bonding bushings.
 2. Protective type clamps.
 3. Bolted type conductor connectors.
 4. Thermal welded type conductor connectors.
 5. Bonding jumpers, straps.
 6. Pressure wire connectors.
 7. Connectors for conductor to ground rod.

3. **EXECUTION**

3.1 Installation – General

1. Install complete permanent, continuous grounding system including, electrodes, conductors, connectors, accessories. Where EMT is used, run ground wire in conduit.
2. Install connectors in accordance with manufacturer's instructions.
3. Protect exposed grounding conductors from mechanical injury with rigid steel conduit as required.
4. Make buried connections, and connections to conductive water main, electrodes, using permanent mechanical connectors or inspectable wrought copper compression connectors to ANSI/IEEE 837.
5. Use mechanical connectors for grounding connections to equipment provided with lugs.
6. Soldered joints not permitted.
7. Install bonding wire for flexible conduit, connected at both ends to grounding bushing, solderless lug, clamp or cup washer and screw. Neatly cleat bonding wire to exterior of flexible conduit.
8. Install flexible ground straps for bus duct enclosure joints, where such bonding is not inherently provided with equipment.
9. Install separate ground conductor to outdoor lighting standards.
10. Install grounding resistance bank if required by contractor provided ground simulation.
11. Ensure the resistance to ground, step, and touch voltages are within permissible levels as per the contractor furnished ground study.
12. Connect building structural steel and metal siding to ground by welding copper to steel.
13. Make grounding connections in radial configuration only, with connections terminating at single grounding point. Avoid loop connections.
14. Bond single conductor, metallic armoured cables in cabinet at supply end, and provide non-metallic entry plate at load end.
15. Ground secondary service pedestals.
16. Install separate ground conductor in all conduits, including branch circuits.

3.2 Electrodes

1. Make ground connections to continuously conductive underground water pipe on street side of water meter.
2. Install water meter shunt.
3. Install concrete encased electrodes in building foundation footings, with terminal connected to grounding network.
4. Install rod electrodes and make grounding connections.
5. Bond separate, multiple electrodes together.
6. Use size 3/0 AWG copper conductors for connections to electrodes.
7. Make special provision for installing electrodes that will give acceptable resistance to ground value where rock or sand terrain prevails. Ground as indicated.

3.3 System and Circuit Grounding

1. Install system and circuit grounding connections to neutral of secondary 600 V system.

3.4 Equipment Grounding

1. Install grounding connections to typical equipment included in, but not necessarily limited to following list: Service equipment, transformers, switchgear, duct systems, frames of motors, motor control centres, starters, control panels, building steel work, generators, elevators and escalators, distribution panels, outdoor lighting, metallic water main.
2. Provide a grounding conductor from the secondary of every distribution transformer to the grounding system. Ground conductor to be sized and installed in accordance with the Canadian Electrical Code.
3. Provide grounding conductor(s) from all major switchgear to solidly ground the secondary system. This includes equipment located in the main electrical room as well as each sub-electrical room. Grounding conductors are to be sized to Canadian Electrical Code and switchgear manufacturer's requirements.

3.5 Grounding Bus

1. Provide a ground bus in the main electrical room. The ground bus shall consist of a suitable length of 50mm x 6mm copper bus mounted on a 25mm insulating standoff. This shall be drilled and tapped to receive all the grounding conductors indicated and an engraved nameplate or tag shall be installed above or below individual conductors indicating their function.
2. Provide a similar ground bus in each sub-electrical room and major mechanical room. Interconnect to the main ground bus with a 3/0 AWG insulated copper grounding conductor that is typically installed with the power feeders.
3. Provide a similar ground bus in each data and voice equipment room and closet as indicated in "Data & Voice Grounding" clause.
4. Ground items of electrical equipment in the electrical room to the ground bus with individual bare stranded copper connections, size 3/0 AWG or as indicated.
5. Copper or bronze lugs are required for termination of all copper conductors at ground busses.

3.6 Communication Systems

1. Install home run insulated ground conductor in conduit from the building main ground bus as follows:
 1. #1/0 AWG to a ground bus in the main telecommunication equipment room.
 2. #2 AWG to a ground bus in the telecommunication room.

3. #2 AWG to a ground bus in the main telephone equipment room.
4. #2 AWG to a ground bus in each telephone backboard in equipment rooms/closets.
2. Unless otherwise solidly bonded, bond all data and telephone incoming and outgoing steel conduits with insulated 1#12 AWG from the nearest "Communication" ground bus.
- 3.7 Field Quality Control
 1. Perform ground continuity and resistance tests using method appropriate to site conditions and to approval of Engineer and local authority having jurisdiction over installation.
 2. Perform tests before energizing electrical system.
 3. Disconnect ground fault indicator during tests.
- 3.8 Maintenance – Clearances
 1. Provide clearance around systems, equipment and components for observation of operation, review, servicing, maintenance and as recommended by manufacturer and CEC, Part 1.
- 3.9 Cleaning
 1. Proceed in accordance with Section 01 74 11 – Cleaning.
 2. On completion and verification of performance of installation, remove surplus materials, excess materials, rubbish, tools and equipment.

END OF SECTION 26 05 28

1. **GENERAL**

1.1 Section Includes

1. Material and Installation for hangers and supports.

1.2 Related Sections

1. This section of the Specification forms part of the contract documents and is to be read, interpreted and coordinated with all other parts.

1.3 References

1. Canadian Standards Association (CSA International)
2. CSA C22.1 (current edition), Canadian Electrical Code, Part 1.

1.4 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.5 Submittals

1. Submittals: in accordance with Section 01 33 00 – Submittal Procedures.
2. Product Data:
 1. Provide manufacturer's printed product literature, specifications and data sheets, and include product characteristics, performance criteria, physical size, finish and limitations.

1.6 Delivery, Storage and Handling

1. Deliver, store, handle and protect materials in accordance with Section 01 61 00 – Common Product Requirements.
2. Deliver, Store and handle materials in accordance with manufacturer's written instructions.
3. Handle materials with suitable lifting equipment.
4. Store materials in heated, dry, weather-protected enclosure.

2. **PRODUCTS**

2.1 Support Channels

1. U shape, size 41 x 41 mm, 2.5 mm thick, surface mounted, suspended or set in poured concrete walls and ceilings.

3. **EXECUTION**

3.1 Installation

1. Secure equipment to masonry, tile and plaster surfaces with lead anchors.
2. Secure equipment to poured concrete with expandable inserts.
3. Secure surface mounted equipment with twist clip fasteners to inverted T-bar ceilings. Ensure that T-bars are adequately supported to carry weight of equipment specified before installation.

4. Support equipment, conduit or cables using clips, spring loaded bolts, cable clamps designed as accessories to basic channel members.
5. Fasten exposed conduit or cables to building construction or support system using straps.
 1. One-hole steel straps to secure surface conduits and cables 50 mm and smaller.
 2. Two-hole steel straps for conduits and cables larger than 50 mm.
 3. Beam clamps to secure conduit to exposed steel work.
6. Suspended support systems.
 1. Support individual cable or conduit runs with 6 mm diameter threaded rods and spring clips.
 2. Support 2 or more cables or conduits on channels supported by 6 mm diameter threaded rod hangers where direct fastening to building construction is impractical.
7. For surface mounting of two or more conduits use channels at 1.5m on center spacing.
8. Provide metal brackets, frames, hangers, clamps and related types of support structures where indicated or as required to support conduit and cable runs.
9. Ensure adequate support for raceways and cables dropped vertically to equipment where there is not wall support.
10. Do not use wire lashing or perforated strap to support or secure raceways or cables.
11. Do not use supports or equipment installed for other trades for conduit or cable support except with permission of other trade and approval of Engineer.
12. Install fastenings and supports as required for each type of equipment cables and conduits, and in accordance with manufacturer's installation recommendations.
13. The contractor shall engage a seismic restraint consultant to design and review seismic restraints for all work associated with division 26. the seismic restraint consultant shall supervise their installation and submit the requisite assurances to the local municipal authorities
14. Submit assurance commitment letter from the seismic restraint consultant at the commencement of the project and its completion
15. All recessed lighting fixtures in mechanical grid ceilings (t- bar) shall be restrained using at least two (2) #16 awg stranded stainless steel aircraft cable security bridles per fixture tied to the basic building structure. attach each security cables at ends of each fixture using a further attachment to each corner of the fixture and in such a manner that the fixture cannot fall lower than 12" (300mm) beneath fixture installed height

3.2 Maintenance – Clearances

1. Provide clearance around systems, equipment and components for observation of operation, review, servicing, maintenance and as recommended by manufacturer and CEC, Part 1.

3.3 Cleaning

1. Proceed in accordance with Section 01 74 11 – Cleaning.
2. On completion and verification of performance of installation, remove surplus materials, excess materials, rubbish, tools and equipment.

END OF SECTION 26 05 29

1. GENERAL

1.1 Section Includes

1. Materials and installation for splitters, junctions, pull boxes and cabinets.

1.2 Related Sections

1. This section of specification forms part of the contract documents and is to be read, interpreted and coordinated with all other parts.

1.3 References

1. Canadian Standards Association (CSA International)
 1. CSA C22.1 (current edition), Canadian Electrical Code, Part 1, 20th Edition.
 2. CSA C22.2 No. 76, Splitters.
 3. CSA C22.2 No. 40, (Cutout), Junction and pull boxes.

1.4 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.5 Submittals

1. Submittals: in accordance with Section 01 33 00 – Submittal Procedures
2. Product Data:
 1. Provided manufacturer's printed product literature, specifications and datasheet and include project characteristics, performance criteria, physical size, finish and limitations.

1.6 Delivery, Storage and Handling

1. Deliver, store, handle and protect materials in accordance with Section 01 61 00 – Common Product Requirements.
2. Deliver, store and handle materials in accordance with manufacturer's written instructions.
3. Handle materials with suitable lifting equipment.
4. Store materials in heated, dry, weather-protected enclosure.

2. PRODUCTS

2.1 Splitters

1. Construction: sheet metal enclosure, welded corners and formed hinged cover suitable for locking in closed position.
2. Terminations: main and branch lugs to match required size and number of incoming and outgoing conductors as indicated.
3. Spare Terminals: minimum three spare terminals or lugs on each connection or lug block sized less than 400 A.

2.2 Junction and Pull Boxes

1. Construction-welded steel enclosure.
2. Covers Flush Mounted: 25 mm minimum extension all around.
3. Covers Surface Mounted: screw-on flat covers.

2.3 Cabinets

1. Construction: welded sheet steel hinged door, handle, latch and catch.
2. Type E Empty: flush overlapping sides, mounting as indicated.
3. Type T Terminal: flush overlapping sides mounting as indicated containing 19 mm plywood backboard.

3. EXECUTION

3.1 Splitter Installation

1. Mount plumb, true and square to building lines.
2. Extend splitters full length of equipment arrangement except where indicated otherwise.

3.2 Junction, Pull Boxes and Cabinets Installation

1. Install pull boxes in inconspicuous but accessible locations.
2. Mount cabinets with top not higher than 2 m above finished floor except where indicated otherwise.
3. Install terminal block as indicated in Type T cabinets.
4. Only main junction and pull boxes are indicated. Install additional pull boxes as required by CSA C22.1.

3.3 Identification

1. Identification Labels: size 2 indicating system name, voltage, phase, number of wires, ampacity, or as indicated.

3.4 Maintenance – Clearances

1. Provide clearance around systems, equipment and components for observation of operation, review, servicing, maintenance and as recommended by manufacturer and CEC, Part 1.

3.5 Cleaning

1. Proceed in accordance with Section 01 74 11 – Cleaning.
 1. On completion and verification of performance of installation, remove surplus materials, excess materials, rubbish, tools and equipment.

END OF SECTION 26 05 31

1. **GENERAL**

1.1 Section includes

1. Materials and installation for outlet boxes, conduit boxes and fittings.

1.2 Related Sections

1. This section of the Specification forms part of the contract documents and is to be read, interpreted and coordinated with all other parts.

1.3 References

1. Canadian Standards Association (CSA International)
 1. CSA C22.1 (current edition), Canadian Electrical Code, Part 1, 20th Edition.
 2. CSA C22.2 No. 18, Outlet Boxes, conduit boxes and fittings.

1.4 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.5 Submittals

1. Submittals: in accordance with Section 01 33 00 – Submittal Procedures.
2. Product Data:
 1. Provide manufacturer's printed project literature, specifications and data sheets, and include product characteristics, performance criteria, physical sizes, finish and limitations.

1.6 Delivery, Storage and Handling

1. Deliver, store, handle and protect materials in accordance with Section 01 61 00 – Common Product Requirements.
2. Deliver, store and handle materials in accordance with manufacturer's written instructions.
3. Handle materials with suitable lifting equipment.
4. Store materials in heated, dry, weather-protected enclosure.

2. **PRODUCTS**

2.1 Outlet and Conduit Boxes – General

1. Size boxes in accordance with CSA C22.1
2. 102 mm square or larger outlet boxes as required.
3. Gang boxes where wiring devices are grouped.
4. Blank cover plates for boxes without wiring devices.
5. Combination boxes with barriers where outlets for more than one system are grouped.

2.2 Galvanized Steel Outlet Boxes

1. One-piece electro-galvanized construction.

2. Single and multi-gang flush device boxes for flush installation, minimum size 76 x 50 x 38 mm or as indicated. 102 mm square outlet boxes when more than one conduit enters one side with extension and plaster rings as required.
3. Utility boxes for outlets connected to surface-mounted EMT conduit, minimum size 102 x 54 x 48 mm.
4. 102 mm square or octagonal outlet boxes for lighting fixture outlets.
5. Extension and plaster rings for flush mounting devices in finished walls.

2.3 Floor Boxes

1. Concrete tight electro-galvanized sheet steel floor boxes with adjustable finishing rings to suit floor finish with brushed aluminum faceplate. Device mounting plate to accommodate short or long ear duplex receptacles. Minimum depth: 73 mm for receptacles and communication outlets.
2. Adjustable, watertight, concrete tight, cast floor boxes with openings drilled and tapped for 16, 21, 27 and 35 mm conduit. Minimum size: 73 mm deep.

2.4 Outlet Boxes for Non-Metallic Sheathed Cable

1. Electro-galvanized, sectional, screw ganging steel boxes, minimum size 76 x 50 x 63 mm with two double clamps to take non-metallic sheathed cables.

2.5 Fittings – General

1. Bushing and connectors with nylon insulated throats.
2. Knock-out fillers to prevent entry of debris.
3. Conduit outlet bodies for conduit up to 35 mm and pull boxes for larger conduits.
4. Double locknuts and insulated bushings on sheet metal boxes.

2.6 Service Fittings

1. 'High Tension' receptacles fitting made of 2-piece die-cast aluminum with brushed aluminum housing finish for two duplex receptacles. Bottom plate with two knockouts for centered or offset installation. 12 x 102 mm extension piece as indicated.
2. Pedestal type 'low tension' fitting made of 2-piece die cast aluminum with brushed aluminum housing finish to accommodate two amphenol jack connectors.

3. **EXECUTION**

3.1 Installation

1. Support boxes independently of connecting conduits.
2. Coordinate locations of boxes and devices with other trades to avoid conflicts with door swings, ceiling tiles, mechanical ducts, pipes, and equipment, and other site conditions.
3. Fill boxes with paper, sponges or foam or similar approved material to prevent entry of debris during construction. Remove upon completion of work.
4. Provide blank covers or plates over all boxes that do not contain devices.
5. For flush installations mount outlets flush with finished wall using plaster rings to permit wall finish to come within 6 mm of opening.
6. Provide correct size of openings in boxes for conduit, mineral insulated and armoured cable connections. Do not install reducing washers.
7. Vacuum clean interior of outlet boxes before installation of wiring devices.

8. Identify systems for outlet boxes as required.
9. The drawings do not necessarily show every junction or pull box; however the Electrical Contractor shall provide all boxes as required.

3.2 Maintenance

1. Provide clearance around systems, equipment and components for observation of operation, review, servicing, maintenance and as recommended by manufacturer and CEC, Part 1.

3.3 Cleaning

1. Proceed in accordance with Section 01 74 11 – Cleaning.
2. On completed and verification of performance of installation, remove surplus materials, excess materials, rubbish, tools and equipment.

END OF SECTION 26 05 32

1. **GENERAL**

1.1 Section Includes

1. Material and installation for conduits, conduit fastenings and conduit fittings.

1.2 Related Sections

1. This section of the Specification forms part of the contract documents and is to be read, interpreted and coordinated with all other parts.

1.3 References

1. Canadian Standard Association (CSA International)
 1. CAN/CSA C22.2 No. 18 (current edition), Outlet Boxes, Conduit Boxes, Fittings and Associated Hardware, A National Standard of Canada.
 2. CSA C22.2 No. 45 (current edition), Rigid Metal Conduit.
 3. CSA C22.2 No. 56 (current edition), Flexible Metal Conduit and Liquid-Tight Flexible Metal Conduit.
 4. CSA C22.2 No. 83 (current edition), Electric Metallic Tubing
 5. CSA C22.2 No. 211.2 (current edition), Rigid PVC (Unplasticized) Conduit.
 6. CAN/CSA C22.2 No. 227.3 (current edition), Nonmetallic Mechanical Protection Tubing (NMPT), A National Standard of Canada (February 2006).

1.4 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.5 Submittals

1. Submittals: in accordance with Section 01 33 00 – Submittal Procedures
2. Product Data:
 1. Provide manufacturer's printed product literature, specifications and data sheets, and include product characteristics, performance criteria, physical size, finish and limitations.

1.6 Delivery, Storage and Handling

1. Deliver, store, handle and protect materials in accordance with Section 01 61 00 – Common Product Requirements.
2. Deliver, store and handle materials in accordance with manufacturer's written instructions.
3. Handle materials with suitable lifting equipment.
4. Store materials with suitable lifting equipment.
5. Store materials in heated, dry, weather-protected enclosure.

2. **PRODUCTS**

2.1 Cables and Reels

1. Provide cables on reels or coils.
 1. Mark or tag each cable and outside of each reel or coil, to indicate cable length, voltage rating, conductor size, and manufacturer's lot number and reel number.
2. Each coil or reel of cable to contain only on continuous cable without splices.

2.2 Conduits

1. Rigid metal conduit: to CSA C22.2 No. 45, hot dipped galvanized steel threaded.
2. Epoxy coated conduit: to CSA C22.2 No. 45, with zinc coating and corrosion resistant epoxy finish inside and outside.
3. Electrical metallic tubing (EMT): to CSA C22.2 No. 83, with couplings.
4. Rigid PVC conduit: to CSA C22.2 No. 211.2.
5. Flexible metal conduit: to CSA C22.2 No. 56, liquid-tight flexible metal.

2.3 Conduit Fastenings

1. One-hole steel straps to secure surface conduits 50 mm and smaller.
 1. Two hole steel straps for conduits larger than 50 mm.
2. Beam clamps to secure conduits to exposed steel work.
3. Channel type supports for two or more conduits at 1.5 m on centre.
4. Threaded rods, 6 mm diameter, to support suspended channels.

2.4 Conduit Fittings

1. Fittings: to CSA C22.2 No. 18, manufactured for use with conduit specified. Coating: same as conduit.
2. Ensure factory "ells" where 90 degrees bends for 25 mm and larger conduits.
3. Watertight connectors and couplings for EMT.
 1. Set-screws are not acceptable.

2.5 Expansion Fittings for Rigid Conduit

1. Weatherproof expansion fittings with internal bonding assembly suitable for 100 mm linear expansion.
2. Watertight expansion fittings with integral bonding jumper suitable for linear expansion and 19 mm deflection.
3. Weatherproof expansion fittings for linear expansion at entry to panel.

2.6 Fish Cord

1. Polypropylene.

3. **EXECUTION**

3.1 Installation

1. Drawings do not show all conduit runs and if shown, are diagrammatic only.
2. Ream and de-burr the ends of each conduit length prior to installation.
3. Install conduits to conserve headroom in exposed locations and cause minimum interference in spaces through which they pass.

4. Conceal conduits except in mechanical and electrical service rooms.
5. Use electrical metallic tubing (EMT) except in cast concrete.
6. Use rigid PVC or DBII conduit underground.
7. Use liquid tight flexible metal conduit for connection to motors or vibrating equipment in damp, wet or corrosive locations.
8. Use explosion proof flexible connection for connection to explosion proof motors.
9. Install conduit sealing fittings in hazardous areas.
 1. Fill with compound.
10. Minimum conduit size for lighting and power circuits: 21 mm.
11. Install EMT conduit from branch circuit panel to outlet boxes located in sub floor.
12. Bend conduit cold:
 1. Replace conduit if kinked or flattened more than 1/10th of its original diameter.
13. Mechanically bend steel conduit over 21 mm diameter.
14. Field threads on rigid conduit must be of sufficient length to draw conduits up tight.
15. Install fish cord in empty conduits.
16. Run two 27 mm spare conduits up to ceiling space and two 27 mm spare conduits down to ceiling space from each flush panel.
 1. Terminate these conduits in 152 x 152 x 102 mm junction boxes in ceiling space or in case of an exposed concrete slab, terminate each conduit in flush concrete type box.
17. Remove and replace blocked conduit sections.
 1. Do not use liquids to clean out conduits.
18. Dry conduits out before installing wire.

3.2 Manufacturer's Instructions

1. Compliance: comply with manufacturer's written recommendations or specifications, including product technical bulletins, handling, storage and installation instructions, and datasheets.

3.3 Surface Conduits

1. Surface conduits shall only be installed in mechanical and electrical rooms.
2. Run parallel or perpendicular to building lines.
3. Locate conduits behind infrared or gas fired heaters with 1.5 m clearance.
4. Run conduits in flanged portion of structural steel.
5. Group conduits wherever possible on suspended channels.
6. Do not pass conduits through structural members except as indicated.
7. Do not locate conduits less than 75 mm parallel to steam or hot water lines with minimum of 25 mm at crossovers.

3.4 Concealed Conduits

1. Run parallel or perpendicular to building lines.

2. Do not install horizontal runs in masonry walls.
3. Do not install conduits in terrazzo or concrete toppings.

3.5 Conduits in Cast-in-Place Concrete

1. Locate to suit reinforcing steel.
 1. Install in center one third of slab.
2. Protect conduits from damage where they stub out of concrete.
3. Install sleeves where conduits pass through slab or wall.
4. Provide oversized sleeve for conduits passing through waterproof membrane, before membrane is installed.
 1. Use cold mastic between sleeve and conduit.
5. Conduits in slabs: minimum slab thickness 4 times conduit diameter.
6. Encase conduits completely in concrete with minimum 25 mm concrete cover.
7. Organize conduits in slab to minimize cross-overs.

3.6 Conduits in Cast-in-Place Slabs on Grade

1. Run conduits 25 mm and larger below slab and encase in 75 mm concrete envelope.
 1. Provide 50 mm of sand over concrete envelope below floor slab.

3.7 Conduits Underground

1. Slope conduits to provide drainage.
2. Waterproof joints (PVC accepted) with heavy coat of bituminous paint.

3.8 Maintenance – Clearances

1. Provide clearance around systems, equipment and components for observation of operation, review, servicing, maintenance and as recommended by manufacturer and CEC, Part 1.

3.9 Cleaning

1. Proceed in accordance with Section 01 74 11 – Cleaning.
2. On completion and verification of performance of installation, remove surplus materials, excess materials, rubbish, tools and equipment.

END OF SECTION 26 05 34

1. **GENERAL**

1.1 Section Includes

1. Materials and installation for Installation of Cables in Trenches and Ducts.

1.2 Related Sections

1. This section of the specification forms part of the contract documents and is to be read, interpreted and coordinated with all other parts.
2. Section 01 74 21 – Construction/Demolition Waste Management and Disposal.
3. Section 31 23 33.01 – Excavating, Trenching and Backfilling.
4. Section 26 05 00 – Common work Results – Electrical.

1.3 References

1. Canadian Standards Association, (CSA International).
 1. CSA C22.1 (current edition), Canadian Electrical Code, Part 1.
2. Insulated Cable Engineers Association, Inc. (ICEA).

1.4 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.5 Submittals

1. Submittals in accordance with Section 01 33 00 – Submittal Procedures.
2. Product Data:
 1. Provide manufacturer's printed product literature, specifications and data sheets, and include product characteristics, performance criteria, physical size, finish and limitations.
3. Sample:
 1. Submit samples in accordance with Section 01 33 00 – Submittal.
4. Closeout Submittals:
 1. Data necessary for maintenance of materials.

1.6 Delivery, Storage and Handling

1. Deliver, store, handle and protect materials in accordance with Section 01 61 00 – Common Product Requirements.
2. Deliver, store and handle materials in accordance with manufacturer's written instructions.
3. Handle materials with suitable lifting equipment.
4. Store materials in heated, dry, weather-protected enclosure.

2. **PRODUCT**

2.1 Markers

1. Concrete type cable markers shall be 600 mm x 600 mm x 100 mm with words, cable, joint or conduit impressed in top surface, with arrows to indicate change in direction of cable and duct runs.

3. **EXECUTION**

3.1 Cable Installation in Ducts

1. Install cables as indicated inside ducts.
 1. Do not pull spliced cables inside ducts.
2. Install multiple cables in duct simultaneously.
3. Use CSA approved lubricants of type compatible with cable jacket to reduce pulling tension.
4. To facilitate matching of colour coded multi-conductor control cables reel off in same direction during installation.
5. Before pulling cables into ducts and until cables are properly terminated, seal ends of lead covered cables with wiping solder; seal ends of non-leaded cables with moisture seal tape.
6. After installation of cables, seal duct ends with duct sealing compound.

3.2 Markers

1. Mark cable every 150 m along duct runs and changes in direction.
2. Mark underground splices.
3. Where markers are removed to permit installation of additional cables, reinstall existing markers.
4. Install cedar post type markers.
5. Lay concrete markers flat and centred over cable with top flush with finish grade.

3.3 Field Quality Control

1. Perform tests in accordance with Section 26 05 00 – Common work Results – Electrical.
2. Perform tests using qualified personnel. Provide necessary instruments and equipment.
3. Check phase rotation and identify each phase conductor of each feeder.
4. Check each feeder for continuity, short circuits and grounds. Ensure resistance to ground of circuits is not less than 50 megohms.
5. Pre-Acceptance Tests:
 1. After installing cable but before splicing and terminating, perform insulation resistance test with 1,000 V megger on each phase conductor.
 2. Check insulation resistance after each splice and/or termination to ensure that cable system is ready for acceptance testing.
6. Acceptance Tests
 1. Ensure that terminations and accessory equipment are disconnected.
 2. Ground shields, ground wires, metallic armour and conductors not under test.
 3. High Potential (Hipot) Testing.
 1. Conduct hipot testing at 133 % of original factory test voltage in accordance with manufacturer's recommendations.

7. Provide Consultant with list of test results showing location at which each test was made, circuit tested and result of each test.
8. Remove and replace entire length of cable if cable fails to meet any of test criteria.

3.4 Maintenance – Clearances

1. Provide clearance around systems, equipment and components for observation of operation, review, servicing, maintenance and as recommended by manufacturer and CEC, Part 1.

3.5 Cleaning

1. Proceed in accordance with Section 01 74 11 – Cleaning
2. On completion and verification of performance of installation, remove surplus materials, excess materials, rubbish, tools and equipment.

END OF SECTION 26 05 43.01

1. GENERAL

1.1 Section Includes

1. Materials and installation for fused and non-fused disconnect switches.

1.2 Related Sections

1. This section of the Specification forms part of the contract documents and is to be read, interpreted and coordinated with all other parts.

1.3 References

1. Canadian Standards Association (CSA International).
 1. CAN/CSA C22.2 No.4 (current edition), Enclosed Switches.
 2. CSA C22.2 No.39 (current edition), Fuseholder Assemblies.

1.4 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.5 Submittals

1. Submit product data in accordance with Section 01 33 00 – Submittal Procedures.
2. Product Data:
 1. Provide manufacturer's printed product literature, specifications and data sheets, and include product characteristics, performance criteria, physical size, finish and limitations.
3. Sample:
 1. Submit samples in accordance with Section 01 33 00 – Submittal Procedures.
4. Closeout Submittals
 1. Provide maintenance data for materials for incorporation into manual specified in Section 01 78 00 – Closeout Submittals.

1.6 Delivery, Storage and Handling

1. Deliver, store, handle and protect materials in accordance with Section 01 61 00 – Common Product Requirements.
2. Deliver, store and handle materials in accordance with manufacturer's written instructions.
3. Handle materials with suitable lifting equipment.
4. Store materials in heated, dry, weather-protected enclosure.

1.7 Acceptable Manufacturers

1. Square D
2. Eaton
3. Or approved equal.

2. PRODUCTS

2.1 Disconnect Switches

1. Fusible, non-fusible, horsepower rated disconnect switch in CSA Enclosure, to CAN/CSA C22.2 No.4 size as indicated.
2. Provision for padlocking in off switch position by locks.
3. Mechanically interlocked door to prevent opening when handle in ON position.
4. Fuseholder: to CSA C22.2 No.39 suitable without adaptors, for type and size of fuse indicated.
5. Quick-make, quick-break action.
6. ON-OFF switch position indication on switch enclosure cover.

3. EXECUTION

3.1 Installation

1. Install disconnect switches complete with fuses if applicable.

3.2 Maintenance – Clearances

1. Provide clearance around systems, equipment and components for observation of operation, review, servicing, maintenance and as recommended by manufacturer and CEC, Part 1.

3.3 Cleaning

1. Proceed in accordance with Section 01 74 11 – Cleaning.
2. On completion and verification of performance of installation, remove surplus materials, excess materials, rubbish, tools and equipment.

END OF SECTION 26 28 23

1. GENERAL

1.1 Section Includes

1. Materials and installation of Surge Protection Devices.

1.2 Related Sections

1. This section of the Specification forms part of the contract documents and is to be read, interpreted and coordinated with all other parts.

1.3 References

1. Canadian Standards Association (CSA International)
 1. CSA C22.1 (current edition), Canadian Electrical Code, Part 1.
2. the specified unit shall be designed, manufacturer, tested and installed in the following standards:
 1. American National Standards Institute of Electrical and Electronic Engineer (ANSI/IEEE C62.41-latest and C62.45-latest).
 2. Canadian Standards Association (CSA) CSA C22.2-Lastest Edition.
 3. Federal Information Processing Standards Publication 94 (FIPS PUB 94)
 4. Underwriters Laboratories (UL 1449, UL 1283).
 5. NEMA LS-1.
 6. ANSI/IEEE C62.41.1 – IEEE Guide for Surge Voltages in Low Voltage AC Power Circuits.
 7. ANSI/IEEE C62.41.2 – IEEE Recommended Practice on Characterization of Surges in Low Voltage (1000 V and under) AC Power Circuits.
 8. ANSI/IEEE C62.45 – IEEE Guide for Surge Testing on Equipment in Low Voltage AC Power Circuits.
 9. FIPS Pub 94 (current edition) – USDC Guide for Installation Techniques on Electrical Power Environments.
 10. NEMA LS-1 Guidebook for the Specification of Surge Protection Devices.
 11. (RoHS) Restriction of Hazardous Substances Directive: All SPDs shall be compliant with the Restriction of Hazardous Substances Directive 2002/95/EC.4.

1.4 Sustainable Requirements

1. Waste Management and Disposal
 1. Separate and recycle waste materials in accordance with local bylaws.
 2. Avoid using landfill waste disposal procedures when recycling facilities are available.
 3. Place materials defined as hazardous or toxic waste in designated containers.

1.5 Submittals

1. Submit in accordance with Section 01 33 00 – Submittal Procedures.
2. Product Data:
 1. Product data shall be submitted in accordance with Section 26 05 00 – Common work Results for Electrical. This shall include schematic diagram and all options such as dry contacts.

2. Provide manufacturer's printed product literature, specifications and data sheets, and include product characteristics, performance criteria, physical size, finish and limitations.
3. Evidence of compliance to the certifications as per CSA and UL 1449 must be This will show actual test data as certified by UL and ANSI standard. Document to include copy of UL Listing Report. Manufacturer shall certify stating that test tests and delivered product both had same suppliers for raw materials and same to manufacture.
4. The following information shall be made available to the Specifying Engineer:
 1. Verification of the SPD Systems' compliance with the required ANSI/UL 1449, 3rd Edition Listing by Underwriter's Laboratories (UL) or other valid, Nationally Recognized Testing Laboratory (NRTL) such as "c ETL us Listed", "c UL us Listed", "c CSA as Listed", "Intertek/ETL LISTED", for engineering review and approval.
 2. Manufacturer's electrical specifications and/or mechanical drawings indicating unit dimensions, weights, full installation instructions, and wiring configuration details.
 3. Compliance data, including Agency Listing or Agency Control Identification Number, Manufacturer's Model Numbers, SPD Type, System Voltage Type, Wiring Diagrams, Voltage Protection Ratings, applicable Technical Specification and/or Mechanical Drawings, and Installation Sheets.
5. Sample:
 1. Submit samples in accordance with Section 01 33 00 – Submittal Procedures.
6. Closeout Submittals:
 1. Provide maintenance data for incorporation into manual specified into manual specified in Section 01 78 00 – Closeout Submittals.

1.6 Delivery, Storage and Handling

1. Deliver, store, handle and protect materials in accordance with Section 01 61 00 – Common Product Requirements.
2. Deliver, store and handle materials in accordance with manufacturer's written instructions.
3. Handle materials with suitable lifting equipment.
4. Store materials in heated, dry, weather-protected enclosure.

1.7 Description of System

1. Formally known as Transient Voltage Surge Suppression (TVSS) and now known as Surge Protection Device (SPD).
2. Surge Suppression Device (SPD) shall consist of a high performance designed to provide surge protection and high-frequency electrical noise filtering.
3. The specified unit shall be compatible with non-linear loads and provide hi transient voltage suppression, surge current diversion and high-frequency electrical noise attenuation.
4. The unit shall be connected in parallel with the electrical distribution system.
5. The operation of the unit shall not be affected by or interact with any other reduction device installed on the electrical distribution system.
6. SPD unit shall be connected to the equipment via a circuit breaker from within the equipment in or SPD unit can be serviced. The SPD shall be of modular design.

1.8 Definitions

1. Single Element: The safety-tested implementation of industrial-grade, “extra-large blocks” MOVs close-coupled/connected to each surge-path. This minimum 45 KA-rated single element per surge path architecture maintains a virtually non-degrading technology for long-term SPD system integrity. Single element technology also provides single voltage and “respond time” thresholds to virtually eliminate inherent and random SPD degradation found in conventional designs. Typical SPD designs leave you guessing if they’re still working after any known or unknown surge events.
2. Zoned and Cascading Protection/Isolation System: The safety-tested implementation of “Zoned” Primary and Secondary levels of industrial-grade, “extra-large block” MOVs close-coupled/connected to each surge-path. Through a facility, this maintains a virtual primary level “grid” of non-degrading SPD protection and isolation of all AC circuits and down-line equipment and components. This effectively separates and isolates the most sensitive in-house electronic circuitry from hazards caused by any nearby operating and switching motor loads, permitting even your most critical sensitive equipment to operate with spec-grade power integrity, thereby eliminating failures and costly downtime.
3. SCCR – Short Circuit Current Rating: the proven suitability of an SPD for use on an AC power circuit that is capable of safely delivering not more than a declared symmetrical or available current at a declared voltage during a “short circuit condition.” The critical SCCR “high current” test is a rigid addition to the recently updated ANSI/UL Standard of 1449, 3rd Edition, that includes user-friendly performance testing and product marking requirements. This indicates how safe the SPD is during and immediately after a catastrophic lightning, utility company AC power event.

1.9 Exclusions

1. No user-accessible, integral or external disconnect switch intended for easy removal of the SPD from the power source is permitted, as it introduces a “weak link” into the intended “surge-path” and effectively defeats the intended purpose of the SPD.
2. SPD designs with “User Replaceable Surge Current Diversion Modules: or “plug-in” replaceable SPD modules shall not be permitted.
3. “Balanced Suppression Platform”, conventional multi-element arrays, or “computer-matched MOV array” technology shall not be permitted as a substitute for safety-fused single-element technology.

1.10 Acceptable Manufacturers

1. Total Protection Solutions
2. Or approved equal.

1.11 Warranty

1. Equipment manufacturer shall warrant that all SPD products supplied are fully performance and safety tested, and are free of operational and material defects in workmanship and performance for the lifetime of the installation.

2. **PRODUCTS**

2.1 General

1. The SPD device shall be suited for operation in the following configuration 120/208VAC, 347/600V VAC, 3 pole, wye, 4 wire, electrical configuration. The operating environment classified by IEEE C62.41 as follows:
 1. Main Switchboard, MCC and distribution panels, 347/600V: Category C3.
 2. Panelboards, 120/208V: Category B.
2. Operating temperature range shall be -25 to +55C.

3. Operation shall be reliable in an environment with 5% to 95% non-condensing humidity.
4. The unit shall not generate any audible noise during normal operation.
5. No appreciable magnetic fields shall be generated by the SPD. The unit shall be used directly in Computer Rooms and any location without danger to storage systems or devices.
6. Type 1 or 2: All SPDs shall be labeled as Type 1 or 2, intended for Load or Line side of over-current protection. All suppression components of all protection modes, including N-G, shall be internally fuse-linked to avoid catastrophic failures and weak links in the actual surge-path. Integral fuse links must be employed to safely fail "open", not before the suppression device becomes full saturated, and only then does the fuse open immediately upon the catastrophic failure of the device. No "fused" or "non-fused" disconnect switches shall be permitted within the SPD circuit or the protected surge-path.
7. SCCR Rating: All SPDs shall be labeled with Independent High Current Lab Test "Short Circuit Current Ratings" (SCCR) of 200kAIC per NEC 285.6. typical thermal fuse ratings shall not be acceptable.
8. UL 1449, 3rd Edition: All SPD submittals shall bear either the "c ETL us Listed", "c UL us Listed", or "c CSA us Listed" mark and shall be listed to the most recent 3rd Edition of UL Standard 1449. Note: "Manufactured 'in compliance with' or 'in accordance with' or 'approved' or 'certified'" is not equivalent to actual ETL, UL or CSA listing and does not qualify for the intent of these specifications.
9. Non-degrading, Single-Element Protection: All suppression components shall consist of a minimum of a single 45 kA rated MOV per mode of surge-path protection on the secondary level, and a minimum of a single 80 kA rated MOV per mode of surge-path protection the primary level.
10. Two-Stage Surge-Path Protection: All primary level SPDs connected to switchgear applications rated over 500kW shall employ two composite stages of "dual-mode" surge-path protection rated at a minimum of 335kA per phase, all primary level SPDs connected to switchgear rated up to 2500kW shall employ two composite stages of "quad-mode" surge-path protection rated at a minimum of 665kA per phase.
11. Safe, Seven-Mode Surge-Path Protection: SPD technology shall employ full seven-mode protection paths to protect all L-L, L-N, L-G and N-G for wye systems; L-L and L-G in delta and impedance-grounded wye systems.
12. Dedicated Disconnect Breaker for Power Panel Installation: Where "most effective SPD performance" placement of SPD for "load-side" placement/connection is desired, the SPD shall be "close-coupled" to a dedicated circuit breaker or disconnect within the power panel or switchgear. However, if the SPD is located as a "Type 1, line-side SPD", no breaker or disconnect is required where the desired SPD or connection point is directly into the primary level Main Distribution Panel, Motor Control Centre or Main Switchgear.

2.2 Suppression Category C

1. Protection Modes. Transient voltage surge suppression components shall be p all possible common and normal modes (line to ground, neutral to ground, li line to neutral). There are ten (10) modes of suppression for a 3 phase WYE configuration.
2. Maximum continuous Operation Voltage (MCOV): in accordance with NEMA LS1-19 the maximum continuous operating voltage of the SPD unit shall be not less than the facility nominal operation voltage.
3. Surge Current: The maximum peak surge current capacity per phase of the spd based on an 8 x 20 microsecond current waveform (described in ANSI/IEEE C62.41-1) shall not be less than:
 1. L-L: 200,000A.

2. L-N: 200,000A.
3. L-G: 200,000A.
4. N-G: 200,000A
4. Performance Rating: the UL listed performance rating for the unit shall be
 1. L-N: 1200V.
 2. L-G: 1200V.
 3. N-G:1200V.
5. Testing shall be performed in accordance with Category A3, B3 and C3 test described in ANSI/IEEE C62.41-1991 (Guidelines for Surge Voltages in Low Voltage Power Circuits). The test procedure shall be in accordance with UL1449 and ANSI/ C62.45-1992. (Guidelines on Surge Testing for Equipment Connected to Low Voltage Power).
6. The SPD filter shall withstand 5000 sequential impulses using the category waveform as described in ANSI/IEEE C62.41-(1991) and performed in accordance with ANSI/IEEE C62.45-1992. The resultant peak let-through voltage of the last impulse vary from the first impulse by more than + 10%. The interval between impulses shall exceed 30 seconds.
7. Suppression System: the SPD unit shall include a solid-state high performance suppression system, utilizing non-linear voltage dependent metal oxide varistors suppression system's components shall not utilize gas tubes, spark gaps, or silicon avalanche diodes.
8. The unit shall have dry contacts to monitor power ON / OFF and alarm status, wired for easy access and replacement and shall be indicated on the drawing with.
9. SPD3: High-energy secondary and basic primary level protection for large equipment and service entrances. Up to 500 kW systems.
 1. Min. Single Element per Mode Surge Current KVA: 80.
 2. Min. Surge Current per Phase KVA: 160.
 3. Min. Total Surge Capacity (KVA @ 8x20usec pulse):
 1. 1PH – 680.
 2. 3PH Wye – 560.
 3. 3PH Delta – 420.
10. SPD4: Very high-energy primary level protection for service entrances and heavy industrial loads. Up to 1000 kW systems.
 1. Min. Single Element per Mode Surge Current kVA: 80.
 2. Min. Surge Current per Phase kVA: 350.
 3. Min. Total Surge Capacity (KVA @ 8x20usec pulse):
 1. 1PH – 680.
 2. 3PH – 1000.
 3. 3PH – 1000.
11. SPD5: Ultra-high energy primary level protection for very large service entrances or when extreme exterior transient activity is present. Up to 2500 kW systems.
 1. Min. Single Element per Mode Surge Current kVA: 80.

2. Min. Surge Current per Phase kVA: 640.
3. Min. Total Surge Capacity (KVA @ 8x20usec pulse):
 1. 1PH – 1400.
 2. 3PH – 2000.
 3. 3PH – 2000.

2.3 Suppression Category B

1. Protection Modes. Transient voltage surge suppression components shall be all possible common and normal modes (line to ground, neutral to ground, line to line to neutral). There are (10) modes of suppression for a 3-phase WYE Conf.
2. Maximum Continuous Operating Voltage (MCOV): in accordance with NEMA LS1-19 the maximum continuous operation voltage of the SPD unit shall be not less than the facility nominal operating voltage.
3. Surge Current: the maximum peak surge current capacity per phase of the spec based on an 8 x 20 microsecond current waveform (described in ANSI/IEEE C62.41-1 shall not be less than:
 1. L-L: 100,000A.
 2. L-N: 100,000A.
 3. L-G: 100,000A.
 4. N-G: 100,000A.
4. Performance Rating: The UL listed performance rating for the unit shall be
 1. L-N: 600V.
 2. L-G: 600V.
 3. N-G: 600V.
5. Testing shall be performed in accordance with Category A3, B3 and C1 test described in ANSI/IEEE C62.41-1991 (Guidelines for Surge Voltages in Low Voltage Power Circuits). The test procedures shall be in accordance with UL1449 and ANSI/IEEE C62.45-1992 (Guidelines on Surge Testing for Equipment Connected to Low Voltage Power).
6. The SPD filter shall withstand 3000 sequential impulses using the category and 8x20 s waveform as described in ANSI/IEEE C62.41-(1991) and performed in accordance with ANSI/IEEE C62.45-1992. The resultant peak let-through voltage of impulse shall not vary from the first impulse by more than + 10%. The interval b impulses shall not exceed 30 seconds.
7. Suppression System: The SPD unit shall include a solid-state high performance suppression system, utilizing non-linear voltage dependent metal oxide varistors suppression system's components shall not utilize gas tubes, spark gaps, or silicone avalanche diodes.
8. SPD1: Small commercial service entrances and secondary level protection of distribution sub-panels, lighting systems and HVAC systems. Up to 250 kW systems.
 1. Min. Single Element per Mode Surge Current KVA: 45.
 2. Min Surge Current per Phase KVA:
 1. 1PH – 90.
 2. 3PH – 135.

3. Min. Total Surge Capacity (KVA @ 8.20usec pulse):
 1. 1PH – 150.
 2. 3PH – 315.
 3. 3PH – 210.
9. SPD2: Standard TVSS features plus “Sine Tracking” high frequency filtering to provide ultra pure power for your sensitive instrumentation, life support and automation equipment. Features very low let-through voltages for costly and critical equipment. Up to 250 kW systems.
 1. Min. Single Element per Mode Surge Current KVA: 45.
 2. Min Surge Current per Phase KVA:
 1. 1PH – 90.
 2. 3PH – 135.
 3. Min. Total Surge Capacity (KVA @ 8.20usec pulse):
 1. 1PH – 150.
 2. 3PH Wye – 315.

2.4 Maximum Continuous Operating Voltage

1. MCOV ratings shall be approximately 115% 9+/- 10% variance) of the Nominal Line to Line, Line to Neutral or Neutral to Ground voltages.

UL Std. 1449, 3rd Ed Voltage Protection Ratings (VPRs)

Nominal Voltage	Line-Line (L-L)	Line-Neutral (L-N)	Neutral-Ground (N-G)
120/240 1 Ph	800	400	800
120/208 Y	800	400	800
240 Delta	800	N/A	N/A
120/240 Hi Leg	800	400	800
480 Delta	1500	900	1500
600 Delta	1800	N/A	N/A
347/600 Y	1800	1000	1800

2.5 Filtering

1. The unit shall include a high-frequency extended range tracking filter. The fast rise-time, high-frequency, transients and electrical line noise attenuation minimum of 60dB at 100 kHz to 100 MHz based on actual insertion loss data obtain utilizing the MIL-STD-220A, 50 ohm insertion loss methodology.

2.6 Enclosures

1. Acceptable enclosures shall be rated under enclosure type NEMA 12, NEMA 4 and/or NEMA 4X (see NEMA Enclosure Ratings).

2.7 Features

1. Internal Connections: All surge current diversion connections shall be a by w impedance wiring. No circuit boards shall be used in transient energy paths.
2. SPD/Filter system connections. No plug-in component modules, quick-disconnect terminals or printed circuit board shall be used in surge current carrying paths.
3. Unit Status indicators: Red status indicator LEDS shall be provided on the cover to indicate the protection status on all phases.

3. EXECUTION

3.1 Installation

1. Integral SPD shall be installed by panel manufacturer/supplier in all equipment SPD units to circuit breaker within the equipment noted on drawings. Conductors as short and straight as possible as, no greater than 18" in length. Input conductor to SPD/filter shall be twisted to reduce impedance during high frequency filtering shall be minimum #6 AWG.
2. The SPD should be installed following the manufacturer's recommended practices and in compliance with all applicable codes.

3.2 Equipment Identification

1. Provide equipment identification nameplate.

3.3 Field Testing

1. Provide complete verification per manufacturer instruction including suggested values on bolted connections.
2. Ensure satisfactory performance to stated standards and if any module or un comply replace it completely.
3. Record insulation resistance from phase to ground according to manufacturer or industry standard. Provide formal records to Engineer for review.

3.4 Maintenance – Clearances

1. Provide clearances around systems, equipment and components for observation of operation, review, servicing, maintenance and as recommended by manufacturer and CEC, Part 1.

3.5 Cleaning

1. Proceed in accordance with Section 01 74 11 – Cleaning
2. On completion and verification of performance of installation, remove surplus materials, excess materials, rubbish, tools and equipment.

3.6 Acceptable Manufacturers

1. Total Protection Solutions
2. Or Engineer approved equivalent

END OF SECTION 26 43 13

Part 1 General

1.1 RELATED REQUIREMENTS

- .1 Section 01 35 43.
- .2 Section 01 45 00.
- .3 Section 01 74 11.
- .4 Section 01 74 21.
- .5 Section 01 77 00.
- .6 Section 01 78 00.
- .7 Section 01 35 29.06

1.2 REFERENCES

- .1 Not Used.

1.3 ADMINISTRATIVE

- .1 Submit to Consultant submittals listed for review. Submit promptly and in orderly sequence to not cause delay in Work. Failure to submit in ample time is not considered sufficient reason for extension of Contract Time and no claim for extension by reason of such default will be allowed.
- .2 Do not proceed with Work affected by submittal until review is complete.
- .3 Present shop drawings, product data, samples and mock-ups in SI Metric units.
- .4 Where items or information is not produced in SI Metric units converted values are acceptable.
- .5 Review submittals prior to submission to Consultant. This review represents that necessary requirements have been determined and verified, or will be, and that each submittal has been checked and co-ordinated with requirements of Work and Contract Documents. Submittals not stamped, signed, dated and identified as to specific project will be returned without being examined and considered rejected.
- .6 Notify Consultant, in writing at time of submission, identifying deviations from requirements of Contract Documents stating reasons for deviations.
- .7 Verify field measurements and affected adjacent Work are co-ordinated.
- .8 Contractor's responsibility for errors and omissions in submission is not relieved by Consultant's review of submittals.
- .9 Contractor's responsibility for deviations in submission from requirements of Contract Documents is not relieved Consultant review.
- .10 Keep one reviewed copy of each submission on site.

1.4 SHOP DRAWINGS AND PRODUCT DATA

- .1 Refer to CCDC 2 GC 3.11.
- .2 The term "shop drawings" means drawings, diagrams, illustrations, schedules, performance charts, brochures and other data which are to be provided by Contractor to illustrate details of a portion of Work.
- .3 Submit drawings stamped and signed by professional engineer registered or licensed in Province of British Columbia.
- .4 Indicate materials, methods of construction and attachment or anchorage, erection diagrams, connections, explanatory notes and other information necessary for completion of Work. Where articles or equipment attach or connect to other articles or equipment, indicate that such items have been co-ordinated, regardless of Section under which adjacent items will be supplied and installed. Indicate cross references to design drawings and specifications.
- .5 Allow 7 days for Consultant's review of each submission.
- .6 Adjustments made on shop drawings by Consultant are not intended to change Contract Price. If adjustments affect value of Work, state such in writing to Consultant prior to proceeding with Work.
- .7 Make changes in shop drawings as Consultant may require, consistent with Contract Documents. When resubmitting, notify Consultant in writing of revisions other than those requested.
- .8 Accompany submissions with transmittal letter, containing:
 - .1 Date.
 - .2 Project title and number.
 - .3 Contractor's name and address.
 - .4 Identification and quantity of each shop drawing, product data and sample.
 - .5 Other pertinent data.
- .9 Submissions include:
 - .1 Date and revision dates.
 - .2 Project title and number.
 - .3 Name and address of:
 - .1 Subcontractor.
 - .2 Supplier.
 - .3 Manufacturer.
 - .4 Contractor's stamp, signed by Contractor's authorized representative certifying approval of submissions, verification of field measurements and compliance with Contract Documents.
 - .5 Details of appropriate portions of Work as applicable:
 - .1 Fabrication.

- .2 Layout, showing dimensions, including identified field dimensions, and clearances.
 - .3 Setting or erection details.
 - .4 Capacities.
 - .5 Performance characteristics.
 - .6 Standards.
 - .7 Operating weight.
 - .8 Wiring diagrams.
 - .9 Single line and schematic diagrams.
 - .10 Relationship to adjacent work.
- .10 After Consultant's review, distribute copies.
- .11 Submit electronic copy of shop drawings for each requirement requested in specification Sections and as Consultant may reasonably request.
- .12 Submit electronic copies of product data sheets or brochures for requirements requested in specification Sections and as requested by Consultant where shop drawings will not be prepared due to standardized manufacture of product.
- .13 Submit electronic copies of test reports for requirements requested in specification Sections and as requested by Consultant.
 - .1 Report signed by authorized official of testing laboratory that material, product or system identical to material, product or system to be provided has been tested in accord with specified requirements.
 - .2 Testing must have been within 1 year of date of contract award for project.
- .14 Submit electronic copies of certificates for requirements requested in specification Sections and as requested Consultant.
 - .1 Statements printed on manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that product, system or material meets specification requirements.
 - .2 Certificates must be dated after award of project contract complete with project name.
- .15 Submit electronic copies of manufacturers instructions for requirements requested in specification Sections and as requested Consultant.
 - .1 Pre-printed material describing installation of product, system or material, including special notices and Material Safety Data Sheets concerning impedances, hazards and safety precautions.
- .16 Submit electronic copies of Manufacturer's Field Reports for requirements requested in specification Sections and as requested by Consultant.
- .17 Documentation of the testing and verification actions taken by manufacturer's representative to confirm compliance with manufacturer's standards or instructions.

- .18 Submit 2 hard copies and 3 electronic copies on thumb drives of Operation and Maintenance Data for requirements requested in specification Sections and as requested by Consultant.
- .19 Delete information not applicable to project.
- .20 Supplement standard information to provide details applicable to project.
- .21 If upon review by Consultant, no errors or omissions are discovered or if only minor corrections are made, an electronic copy will be returned and fabrication and installation of Work may proceed. If shop drawings are rejected, noted copy will be returned and resubmission of corrected shop drawings, through same procedure indicated above, must be performed before fabrication and installation of Work may proceed.
- .22 The review of shop drawings by Public Works and Government Services Canada (PWGSC) is for sole purpose of ascertaining conformance with general concept.
 - .1 This review shall not mean that PWGSC approves detail design inherent in shop drawings, responsibility for which shall remain with Contractor submitting same, and such review shall not relieve Contractor of responsibility for errors or omissions in shop drawings or of responsibility for meeting requirements of construction and Contract Documents.
 - .2 Without restricting generality of foregoing, Contractor is responsible for dimensions to be confirmed and correlated at job site, for information that pertains solely to fabrication processes or to techniques of construction and installation and for co-ordination of Work of sub-trades.

1.5 SAMPLES

- .1 Submit for review samples as requested in respective specification Sections. Label samples with origin and intended use.
- .2 Deliver samples prepaid to the project address.
- .3 Notify Consultant in writing, at time of submission of deviations in samples from requirements of Contract Documents.
- .4 Where colour, pattern or texture is criterion, submit full range of samples.
- .5 Adjustments made on samples by Consultant are not intended to change Contract Price. If adjustments affect value of Work, state such in writing to Consultant prior to proceeding with Work.
- .6 Make changes in samples which Consultant may require, consistent with Contract Documents.
- .7 Reviewed and accepted samples will become standard of workmanship and material against which installed Work will be verified.

1.6 MOCK-UPS

- .1 Erect mock-ups in accordance with 01 45 00 - Quality Control.

1.7 PHOTOGRAPHIC DOCUMENTATION

- .1 Submit electronic copy of colour] digital photography in jpg format, standard resolution monthly with progress statement as directed by Consultant.
- .2 Project identification: name and number of project and date of exposure indicated.
- .3 Frequency of photographic documentation: monthly as directed by Consultant.
 - .1 Upon completion of: excavation, of Work and as directed Consultant.

1.8 CERTIFICATES AND TRANSCRIPTS

- .1 Immediately after award of Contract, submit WorkSafe BC status.
- .2 Submit transcription of insurance immediately after award of Contract.

Part 2 Products

2.1 NOT USED

- .1 Not Used.

Part 3 Execution

3.1 NOT USED

- .1 Not Used.

END OF SECTION

Part 1 General

1.1 RELATED REQUIREMENTS

- .1 Section 01 33 00.
- .2 Section 01 35 43.
- .3 Section 01 45 00.
- .4 Section 01 74 11.
- .5 Section 01 74 21.
- .6 Section 01 77 00.
- .7 Section 01 78 00.

1.2 REFERENCES

- .1 Canada Labour Code, Part 2, Canada Occupational Safety and Health Regulations
- .2 Province of British Columbia
 - .1 Workers Compensation Act, RSBC - (current edition).

1.3 ACTION AND INFORMATIONAL SUBMITTALS

- .1 Submit in accordance with Section 01 33 00 - Submittal Procedures.
- .2 Submit site-specific Health and Safety Plan: Within 7 days after date of Notice to Proceed and prior to commencement of Work. Health and Safety Plan must include:
 - .1 Results of site specific safety hazard assessment.
 - .2 Results of safety and health risk or hazard analysis for site tasks and operation found in work plan.
- .3 Submit 1 digital copy of Contractor's authorized representative's work site health and safety inspection reports to authority having jurisdiction, monthly, and to Departmental Representative and Consultant.
- .4 Submit copies of reports or directions issued by Federal, Provincial and Territorial health and safety inspectors.
- .5 Submit copies of incident and accident reports.
- .6 Submit WHMIS MSDS - Material Safety Data Sheets.
- .7 Departmental Representative and Consultant will review Contractor's site-specific Health and Safety Plan and provide comments to Contractor within 10 days after receipt of plan. Revise plan as appropriate and resubmit plan to Departmental Representative and Consultant within 10 days after receipt of comments from Departmental Representative and/or Consultant.
- .8 Departmental Representative's and Consultant's review of Contractor's final Health and Safety plan should not be construed as approval and does not reduce the Contractor's overall responsibility for construction Health and Safety.

- .9 Medical Surveillance: where prescribed by legislation, regulation or safety program, submit certification of medical surveillance for site personnel prior to commencement of Work, and submit additional certifications for any new site personnel to Departmental Representative and Consultant.
- .10 On-site Contingency and Emergency Response Plan: address standard operating procedures to be implemented during emergency situations.

1.4 FILING OF NOTICE

- .1 File Notice of Project with Provincial authorities prior to beginning of Work.
- .2 Contractor shall be responsible and assume the Principal Contractor role for each work zone location and not the entire complex. Contractor shall provide a written acknowledgement of this responsibility with 3 weeks of contract award.
- .3 Work zone locations include:
 - .1 Kitimat River Hatchery.
- .4 Contractor shall agree to install proper site separation and identification in order to maintain time and space at all times throughout life of project.

1.5 SAFETY ASSESSMENT

- .1 Perform site specific safety hazard assessment related to project.

1.6 MEETINGS

- .1 Schedule and administer Health and Safety meeting with Departmental Representative and/or Consultant prior to commencement of Work.

1.7 PROJECT/SITE CONDITIONS

- .1 Work at site will involve contact with:
 - .1 Salmonid Enhancement Program (SEP) employees.
 - .2 Members of the Public.

1.8 GENERAL REQUIREMENTS

- .1 Develop written site-specific Health and Safety Plan based on hazard assessment prior to beginning site Work and continue to implement, maintain, and enforce plan until final demobilization from site. Health and Safety Plan must address project specifications.
- .2 Departmental Representative and/or Consultant may respond in writing, where deficiencies or concerns are noted and may request re-submission with correction of deficiencies or concerns.

1.9 RESPONSIBILITY

- .1 Be responsible for health and safety of persons on site, safety of property on site and for protection of persons adjacent to site and environment to extent that they may be affected by conduct of Work.

- .2 Comply with and enforce compliance by employees with safety requirements of Contract Documents, applicable federal, provincial, territorial and local statutes, regulations, and ordinances, and with site-specific Health and Safety Plan.

1.10 COMPLIANCE REQUIREMENTS

- .1 Comply with Workers Compensation Act, B.C.
- .2 Comply with Canada Labour Code, Canada Occupational Safety and Health Regulations.

1.11 UNFORSEEN HAZARDS

- .1 When unforeseen or peculiar safety-related factor, hazard, or condition occur during performance of Work, follow procedures in place for Employee's Right to Refuse Work in accordance with Acts and Regulations of Province having jurisdiction and advise Departmental Representative and Consultant verbally and in writing.
- .2 When unforeseen or peculiar safety-related factor, hazard, or condition occur during performance of Work, advise Health and Safety co-ordinator and/or Safety Officer and follow procedures in accordance with Acts and Regulations of Province having jurisdiction and advise Departmental Representative and Consultant verbally and in writing.

1.12 HEALTH AND SAFETY CO-ORDINATOR

- .1 Employ and assign to Work, competent and authorized representative as Health and Safety Co-ordinator. Health and Safety Co-ordinator must:
 - .1 Have site-related working experience specific to activities associated with electrical upgrades.
 - .2 Have working knowledge of occupational safety and health regulations.
 - .3 Be responsible for completing Contractor's Health and Safety Training Sessions and ensuring that personnel not successfully completing required training are not permitted to enter site to perform Work.
 - .4 Be responsible for implementing, enforcing daily and monitoring site-specific Contractor's Health and Safety Plan.

1.13 POSTING OF DOCUMENTS

- .1 Ensure applicable items, articles, notices and orders are posted in conspicuous location on site in accordance with Acts and Regulations of Province having jurisdiction, and in consultation with Departmental Representative and Consultant.

1.14 CORRECTION OF NON-COMPLIANCE

- .1 Immediately address health and safety non-compliance issues identified by authority having jurisdiction or by Departmental Representative and Consultant.
- .2 Provide Departmental Representative and Consultant with written report of action taken to correct non-compliance of health and safety issues identified.

- .3 Departmental Representative and/or Consultant may stop Work if non-compliance of health and safety regulations is not corrected.

1.15 POWDER ACTUATED DEVICES

- .1 Use powder actuated devices only after receipt of written permission from Departmental Representative and/or Consultant.

1.16 WORK STOPPAGE

- .1 Give precedence to safety and health of public and site personnel and protection of environment over cost and schedule considerations for Work.

Part 2 Products

2.1 NOT USED

- .1 Not used.

Part 3 Execution

3.1 NOT USED

- .1 Not used.

END OF SECTION

Part 1 General

1.1 RELATED REQUIREMENTS

- .1 Section 01 33 00.
- .2 Section 01 45 00.
- .3 Section 01 74 11.
- .4 Section 01 74 21.
- .5 Section 01 77 00.
- .6 Section 01 78 00.
- .7 Section 01 35 29.06

1.2 REFERENCES

- .1 Definitions:
 - .1 Environmental Pollution and Damage: presence of chemical, physical, biological elements or agents which adversely affect human health and welfare; unfavourably alter ecological balances of importance to human life; affect other species of importance to humans; or degrade environment aesthetically, culturally and/or historically.
 - .2 Environmental Protection: prevention/control of pollution and habitat or environment disruption during construction.
- .2 Reference Standards:
 - .1 Canadian Construction Documents Committee (CCDC)
 - .1 CCDC 2-2008 Stipulated Price Contract.

1.3 ACTION AND INFORMATIONAL SUBMITTALS

- .1 Submit in accordance with Section 01 33 00 - Submittal Procedures.
- .2 Product Data:
 - .1 Submit manufacturer's instructions, printed product literature and data sheets for electrical equipment and include product characteristics, performance criteria, physical size, finish and limitations.
 - .2 Submit 1 digital copy of WHMIS MSDS in accordance with Section 01 35 29.06 - Health and Safety Requirements.
- .3 Before commencing construction activities or delivery of materials to site, submit Environmental Protection Plan for review and approval by Departmental Representative and Consultant.
- .4 Environmental Protection Plan must include comprehensive overview of known or potential environmental issues to be addressed during construction.
- .5 Address topics at level of detail commensurate with environmental issue and required construction tasks.

-
- .6 Include in Environmental Protection Plan:
- .1 Names of persons responsible for ensuring adherence to Environmental Protection Plan.
 - .2 Names and qualifications of persons responsible for manifesting hazardous waste to be removed from site.
 - .3 Names and qualifications of persons responsible for training site personnel.
 - .4 Descriptions of environmental protection personnel training program.
 - .5 Erosion and sediment control plan identifying type and location of erosion and sediment controls to be provided including monitoring and reporting requirements to assure that control measures are in compliance with erosion and sediment control plan, Federal, Provincial, and Municipal laws and regulations.
 - .6 Drawings indicating locations of proposed temporary excavations or embankments for haul roads, stream crossings, material storage areas, structures, sanitary facilities, and stockpiles of excess or spoil materials including methods to control runoff and to contain materials on site.
 - .7 Traffic Control Plans including measures to reduce erosion of temporary roadbeds by construction traffic, especially during wet weather.
 - .1 Plans to include measures to minimize amount of material transported onto paved public roads by vehicles or runoff.
 - .8 Work area plan showing proposed activity in each portion of area and identifying areas of limited use or non-use.
 - .1 Plan to include measures for marking limits of use areas and methods for protection of features to be preserved within authorized work areas.
 - .9 Spill Control Plan to include procedures, instructions, and reports to be used in event of unforeseen spill of regulated substance.
 - .10 Non-Hazardous solid waste disposal plan identifying methods and locations for solid waste disposal including clearing debris.
 - .11 Air pollution control plan detailing provisions to assure that dust, debris, materials, and trash, are contained on project site.
 - .12 Contaminant Prevention Plan identifying potentially hazardous substances to be used on job site; intended actions to prevent introduction of such materials into air, water, or ground; and detailing provisions for compliance with Federal, Provincial, and Municipal laws and regulations for storage and handling of these materials.
 - .13 Waste Water Management Plan identifying methods and procedures for management of discharge of waste waters which are directly derived from construction activities, such as concrete curing water, clean-up water, dewatering of ground water, disinfection water, hydrostatic test water, and water used in flushing of lines.
 - .14 Historical, archaeological, cultural resources biological resources and wetlands plan that defines procedures for identifying and protecting historical, archaeological, cultural resources, biological resources and wetlands.

1.4 FIRES

- .1 Fires and burning of rubbish on site is not permitted.
- .2 Provide supervision, attendance and fire protection measures as directed.

1.5 DRAINAGE

- .1 Develop and submit erosion and Sediment Control Plan (SCP) identifying type and location of erosion and sediment controls provided. Plan to include monitoring and reporting requirements to assure that control measures are in compliance with erosion and sediment control plan, Federal, Provincial, and Municipal laws and regulations.
- .2 Storm Water Pollution Prevention Plan (SWPPP) to be substituted for erosion and sediment control plan.
- .3 Provide temporary drainage and pumping required to keep excavations and site free from water.
- .4 Ensure pumped water into waterways, sewer or drainage systems is free of suspended materials.
- .5 Control disposal or runoff of water containing suspended materials or other harmful substances in accordance with local authority requirements.

1.6 SITE CLEARING AND PLANT PROTECTION

- .1 Protect trees and plants on site and adjacent properties as indicated.
- .2 Protect trees and shrubs adjacent to construction work.
- .3 Minimize stripping of topsoil and vegetation.

1.7 WORK ADJACENT TO WATERWAYS

- .1 Construction equipment to be operated on land only.
- .2 Waterways to be kept free of excavated fill, waste material and debris.
- .3 Design and construct temporary crossings to minimize erosion to waterways.
- .4 Do not skid logs or construction materials across waterways.
- .5 Avoid indicated spawning beds when constructing temporary crossings of waterways.

1.8 POLLUTION CONTROL

- .1 Maintain temporary erosion and pollution control features installed under this Contract.
- .2 Control emissions from equipment and plant in accordance with local authorities' emission requirements.
- .3 Prevent sandblasting and other extraneous materials from contaminating air and waterways beyond application area.
- .4 Cover or wet down dry materials and rubbish to prevent blowing dust and debris. Provide dust control for temporary roads.

1.9 HISTORICAL/ARCHAEOLOGICAL CONTROL

- .1 Provide historical, archaeological, cultural resources, biological resources, and wetlands plan that defines procedures for identifying and protecting historical, archaeological, cultural resources, biological resources and wetlands known to be on project site: and identifies procedures to be followed if historical archaeological, cultural resources, biological resources and wetlands not previously known to be onsite or in area are discovered during construction.
- .2 Plan: include methods to assure protection of known or discovered resources and identify lines of communication between Contractor personnel and Departmental Representative & Consultant.

1.10 NOTIFICATION

- .1 Departmental Representative & Consultant will notify Contractor in writing of observed noncompliance with Federal, Provincial or Municipal environmental laws or regulations, permits, and other elements of Contractor's Environmental Protection plan.
- .2 Contractor: after receipt of such notice, inform Departmental Representative & Consultant of proposed corrective action and take such action for approval by Departmental Representative.
 - .1 Take action only after receipt of written approval by Departmental Representative.
- .3 Departmental Representative or Consultant will issue stop order of work until satisfactory corrective action has been taken.
- .4 No time extensions granted or equitable adjustments allowed to Contractor for such suspensions.

Part 2 Products

2.1 NOT USED

- .1 Not Used.

Part 3 Execution

3.1 CLEANING

- .1 Progress Cleaning: clean in accordance with Section 01 74 11 - Cleaning.
 - .1 Leave Work area clean at end of each day.
- .2 Remove rubbish and waste materials from site as directed after receipt of written approval from Departmental Representative or Consultant.
- .3 Ensure public waterways, storm and sanitary sewers remain free of waste and volatile materials disposal.
- .4 Final Cleaning: upon completion remove surplus materials, rubbish, tools and equipment in accordance with Section 01 74 11 - Cleaning.

- .5 Waste Management: separate waste materials for recycling in accordance with Section 01 74 21 - Construction/Demolition Waste Management.
 - .1 Remove recycling containers and bins from site and dispose of materials at appropriate facility.

END OF SECTION

Part 1 General

1.1 RELATED REQUIREMENTS

- .1 Section 01 33 00.
- .2 Section 01 35 43.
- .3 Section 01 74 11.
- .4 Section 01 74 21.
- .5 Section 01 77 00.
- .6 Section 01 78 00.
- .7 Section 01 35 29.06

1.2 REFERENCES

- .1 Canadian Construction Documents Committee (CCDC)
 - .1 CCDC 2, Stipulated Price Contract.

1.3 INSPECTION

- .1 Refer to CCDC 2, GC 2.3.
- .2 Allow Departmental Representative or Consultant access to Work. If part of Work is in preparation at locations other than Place of Work, allow access to such Work whenever it is in progress.
- .3 Give timely notice requesting inspection if Work is designated for special tests, inspections or approvals by Departmental Representative or Consultant instructions, or law of Place of Work.
- .4 If Contractor covers or permits to be covered Work that has been designated for special tests, inspections or approvals before such is made, uncover such Work, have inspections or tests satisfactorily completed and make good such Work.
- .5 Departmental Representative or Consultant will order part of Work to be examined if Work is suspected to be not in accordance with Contract Documents. If, upon examination such work is found not in accordance with Contract Documents, correct such Work and pay cost of examination and correction.

1.4 INDEPENDENT INSPECTION AGENCIES

- .1 Independent Inspection/Testing Agencies will be engaged by Departmental Representative for purpose of inspecting and/or testing portions of Work. Cost of such services will be borne by Departmental Representative.
- .2 Provide equipment required for executing inspection and testing by appointed agencies.
- .3 Employment of inspection/testing agencies does not relax responsibility to perform Work in accordance with Contract Documents.

- .4 If defects are revealed during inspection and/or testing, appointed agency will request additional inspection and/or testing to ascertain full degree of defect. Correct defect and irregularities as advised by Departmental Representative or Consultant at no cost to Departmental Representative. Pay costs for retesting and reinspection.

1.5 ACCESS TO WORK

- .1 Allow inspection/testing agencies access to Work, off site manufacturing and fabrication plants.
- .2 Co-operate to provide reasonable facilities for such access.

1.6 PROCEDURES

- .1 Notify appropriate agency and Departmental Representative & Consultant in advance of requirement for tests, in order that attendance arrangements can be made.
- .2 Submit samples and/or materials required for testing, as specifically requested in specifications. Submit with reasonable promptness and in orderly sequence to not cause delays in Work.
- .3 Provide labour and facilities to obtain and handle samples and materials on site. Provide sufficient space to store and cure test samples.

1.7 REJECTED WORK

- .1 Refer to CCDC, GC 2.4.
- .2 Remove defective Work, whether result of poor workmanship, use of defective products or damage and whether incorporated in Work or not, which has been rejected by Departmental Representative or Consultant as failing to conform to Contract Documents. Replace or re-execute in accordance with Contract Documents.
- .3 Make good other Contractor's work damaged by such removals or replacements promptly.
- .4 If in opinion of Departmental Representative or Consultant, it is not expedient to correct defective Work or Work not performed in accordance with Contract Documents, Owner will deduct from Contract Price difference in value between Work performed and that called for by Contract Documents, amount of which will be determined by .

1.8 REPORTS

- .1 Submit 1 digital copy of inspection and test reports to Departmental and Consultant.
- .2 Provide copies to subcontractor of work being inspected or tested, manufacturer or fabricator of material being inspected or tested.

1.9 TESTS AND MIX DESIGNS

- .1 Furnish test results and mix designs as requested.

- .2 Cost of tests and mix designs beyond those called for in Contract Documents or beyond those required by law of Place of Work will be appraised by Departmental Representative or Consultant and may be authorized as recoverable.

1.10 MOCK-UPS

- .1 Prepare mock-ups for Work specifically requested in specifications. Include for Work of Sections required to provide mock-ups.
- .2 Construct in locations as specified in specific Section or acceptable to Departmental Representative or Consultant.
- .3 Prepare mock-ups for Departmental Representative or Consultant's review with reasonable promptness and in orderly sequence, to not cause delays in Work.
- .4 Failure to prepare mock-ups in ample time is not considered sufficient reason for extension of Contract Time and no claim for extension by reason of such default will be allowed.
- .5 If requested, Departmental Representative or Consultant will assist in preparing schedule fixing dates for preparation.
- .6 Remove mock-up at conclusion of Work or when acceptable to Departmental Representative or Consultant.
- .7 Mock-ups may remain as part of Work.
- .8 Specification section identifies whether mock-up may remain as part of Work or if it is to be removed and when.

1.11 EQUIPMENT AND SYSTEMS

- .1 Submit adjustment and testing reports for electrical and building equipment systems.
- .2 Refer to Electrical Services Upgrade Drawing for definitive requirements.

Part 2 Products

2.1 NOT USED

- .1 Not Used.

Part 3 Execution

3.1 NOT USED

- .1 Not Used.

END OF SECTION

Part 1 General

1.1 RELATED REQUIREMENTS

- .1 Section 01 33 00.
- .2 Section 01 35 43.
- .3 Section 01 45 00.
- .4 Section 01 74 21.
- .5 Section 01 77 00.
- .6 Section 01 78 00.
- .7 Section 01 35 29.06

1.2 REFERENCES

- .1 Canadian Construction Documents Committee (CCDC)
 - .1 CCDC 2, Stipulated Price Contract.

1.3 PROJECT CLEANLINESS

- .1 Maintain Work in tidy condition, free from accumulation of waste products and debris.
- .2 Remove waste materials from site at daily regularly scheduled times or dispose of as directed by Departmental Representative or Consultant. Do not burn waste materials on site.
- .3 Make arrangements with and obtain permits from authorities having jurisdiction for disposal of waste and debris.
- .4 Provide on-site recycle and debris containers for collection of waste materials and debris.
- .5 Provide and use marked separate bins for recycling. Refer to Section 01 74 21 - Construction/Demolition Waste Management and Disposal.
- .6 Dispose of waste materials and debris off site.
- .7 Clean interior areas prior to start of finishing work, and maintain areas free of dust and other contaminants during finishing operations.
- .8 Store volatile waste in covered metal containers, and remove from premises at end of each working day.
- .9 Provide adequate ventilation during use of volatile or noxious substances. Use of building ventilation systems is not permitted for this purpose.
- .10 Use only cleaning materials recommended by manufacturer of surface to be cleaned, and as recommended by cleaning material manufacturer.
- .11 Schedule cleaning operations so that resulting dust, debris and other contaminants will not fall on wet, newly painted surfaces nor contaminate building systems.

1.4 FINAL CLEANING

- .1 Refer to CCDC 2, GC 3.14.
- .2 When Work is Substantially Performed remove surplus products, tools, construction machinery and equipment not required for performance of remaining Work.
- .3 Remove waste products and debris other than that caused by others, and leave Work clean and suitable for occupancy.
- .4 Prior to final review remove surplus products, tools, construction machinery and equipment.
- .5 Remove waste materials from site at regularly scheduled times or dispose of as directed by Departmental Representative or Consultant. Do not burn waste materials on site.
- .6 Make arrangements with and obtain permits from authorities having jurisdiction for disposal of waste and debris.
- .7 Remove stains, spots, marks and dirt from all work and electrical equipment.
- .8 Vacuum clean and dust building interiors, behind grilles, louvres and screens.
- .9 Inspect finishes, fitments and equipment and ensure specified workmanship and operation.
- .10 Broom clean and wash exterior walks, steps and surfaces; rake clean other surfaces of grounds.
- .11 Remove dirt and other disfiguration from exterior surfaces.
- .12 Sweep and wash clean paved areas.
- .13 Remove debris and surplus materials from crawl areas and other accessible concealed spaces.

1.5 WASTE MANAGEMENT AND DISPOSAL

- .1 Separate waste materials for recycling in accordance with Section 01 74 21 - Construction/Demolition Waste Management And Disposal.

Part 2 Products

2.1 NOT USED

- .1 Not Used.

Part 3 Execution

3.1 NOT USED

- .1 Not Used.

END OF SECTION

Part 1 General

1.1 WASTE MANAGEMENT GOALS

- .1 Prior to start of Work conduct meeting with Departmental Representative and Consultant to review and discuss PWGSC's waste management goal and Contractor's proposed Waste Reduction Workplan for Construction, Renovation and /or Demolition (CRD) waste to be project generated.
- .2 PWGSC's waste management goal: to divert a minimum 75 percent of total Project Waste from landfill sites. Prior to project completion provide Departmental Representative & Consultant documentation certifying that waste management, recycling, reuse of recyclable and reusable materials have been extensively practiced. The overall waste diversion goal for this project is 75%.
- .3 Specific material target percentages for reuse and/or recycling:
 - .1 Masonry and pavement: 75%.
 - .2 Ceilings and walls: 75%.
 - .3 Metals: 75%.
 - .4 Wood: 75%.
 - .5 Electrical - wiring/conduits/boxes: 75%.
 - .6 Electrical - lighting: 75%.
 - .7 Electrical - other: 75%.
 - .8 Packaging: 75%.
- .4 Target percentage goals are achievable for waste diversion. Contractor to review and confirm Departmental Representative's & Consultant's Waste Audit acceptable values.
- .5 Minimize amount of non-hazardous solid waste generated by project and accomplish maximum source reduction, reuse and recycling of solid waste produced by CRD activities.
- .6 Protect environment and prevent environmental pollution damage.

1.2 RELATED REQUIREMENTS

- .1 Section 01 33 00.
- .2 Section 01 35 43.
- .3 Section 01 45 00.
- .4 Section 01 74 11.
- .5 Section 01 77 00.
- .6 Section 01 78 00.
- .7 Section 01 35 29.06

1.3 REFERENCES

.1 Definitions:

- .1 Approved/Authorized recycling facility: waste recycler approved by applicable provincial authority or other users of material for recycling approved by the Departmental Representative.
- .2 Class III: non-hazardous waste - construction renovation and demolition waste.
- .3 Construction, Renovation and/or Demolition (CRD) Waste: Class III solid, non-hazardous waste materials generated during construction, demolition, and/or renovation activities
- .4 Cost/Revenue Analysis Workplan (CRAW): based on information from Waste Reduction Workplan, and intended as financial tracking tool for determining economic status of waste management practices (Schedule E).
- .5 Inert Fill: inert waste - exclusively asphalt and concrete.
- .6 Waste Source Separation Program (WSSP): implementation and co-ordination of ongoing activities to ensure designated waste materials will be sorted into pre-defined categories and sent for recycling and reuse, maximizing diversion and potential to reduce disposal costs.
- .7 Recyclable: ability of product or material to be recovered at end of its life cycle and re-manufactured into new product for reuse.
- .8 Recycle: process by which waste and recyclable materials are transformed or collected for purpose of being transferred into new products.
- .9 Recycling: process of sorting, cleansing, treating and reconstituting solid waste and other discarded materials for purpose of using in altered form. Recycling does not include burning, incinerating, or thermally destroying waste.
- .10 Salvage: removal of structural and non-structural materials from deconstruction/disassembly projects for purpose of reuse or recycling.
- .11 Separate Condition: refers to waste sorted into individual types.
- .12 Source Separation: act of keeping different types of waste materials separate beginning from the point they became waste.
- .13 Waste Audit (WA): detailed inventory of estimated quantities of waste materials that will be generated during construction, demolition, deconstruction and/or renovation. Involves quantifying by volume/weight amounts of materials and wastes that will be reused, recycled or landfilled. Refer to Schedule A.
- .14 Waste Diversion Report: detailed report of final results, quantifying cumulative weights and percentages of waste materials reused, recycled and landfilled over course of project. Measures success against Waste Reduction Workplan (WRW) goals and identifies lessons learned.
- .15 Waste Management Co-ordinator (WMC) : contractor representative responsible for supervising waste management activities as well as co-ordinating required submittal and reporting requirements.

- .16 Waste Reduction Workplan (WRW): written report which addresses opportunities for reduction, reuse, or recycling of materials generated by project. Specifies diversion goals, implementation and reporting procedures, anticipated results and responsibilities. Waste Reduction Workplan (Schedule B) information acquired from Waste Audit.
- .2 Reference Standards:
 - .1 Canada Green Building Council (CaGBC)
 - .1 LEED Canada for Existing Buildings, Operations and Maintenance, LEED Canada (current edition) Leadership In Energy and Environmental Design Green Building Rating System Reference Guide.
 - .2 Ontario Ministry of Environment
 - .1 Canadian Construction Association (CCA)
 - .1 CCA 81- (current edition): A Best Practices Guide to Solid Waste Reduction.
 - .2 Public Works and Government Services Canada (PWGSC)
 - .1 2002 National Construction, Renovation and Demolition Non-Hazardous Solid Waste Management Protocol.
 - .2 CRD Waste Management Market Research Report (available from PWGSC's Environmental Services).
 - .3 Sustainable Development Strategy (current edition): Target 2.1 Environmentally Sustainable Use of Natural Resources.
 - .1 Real Property projects over \$1 million and in communities where industrial recycling is supported, implementation of CRD waste management practices will be completed, with waste materials being reused or recycled.
 - .2 Contractually ensure resources used in construction or maintenance are consumed and recovered in a sustainable manner.

1.4 DOCUMENTS

- .1 Post and maintain in visible and accessible area at job site, one copy of following documents:
 - .1 Waste Audit (Schedule A).
 - .2 Waste Reduction Workplan (Schedule B).
 - .3 Waste Source Separation Program.
 - .4 Schedules A and B completed for project.

1.5 ACTION AND INFORMATIONAL SUBMITTALS

- .1 Submit in accordance with Section 01 33 00 - Submittal Procedures.
- .2 Prepare and submit following prior to project start-up:

- .1 1 electronic copy of completed Waste Audit (WA): Schedule A.
- .2 1 electronic copy of completed Waste Reduction Workplan (WRW): Schedule B.
- .3 1 electronic copy of Cost/Revenue Analysis Workplan (CRAW): Schedule E.
- .4 1 electronic copy of Waste Source Separation Program (WSSP).
- .3 Prepare and submit on monthly basis, throughout project or at intervals agreed to by Departmental Representative & Consultant the following:
 - .1 Receipts, scale tickets, waybills, and/or waste disposal receipts that show quantities and types of materials reused, recycled, or disposed of.
 - .2 Updated Waste Materials Tracking form (Schedule D).
 - .3 Written monthly summary report detailing cumulative amounts of waste materials reused, recycled and landfilled, and brief status of ongoing waste management activities.
- .4 Submit prior to final payment the following:
 - .1 Waste Diversion Report, indicating final quantities by material types salvaged for reuse, recycling or disposal in landfill and recycling centres, re-use depots, landfills and other waste processors that received waste materials (See Schedule C).
 - .2 Provide receipts, scale tickets, waybills, waste disposal receipts that confirm quantities and types of materials reused, recycled or disposed of and destination.

1.6 WASTE AUDIT (WA)

- .1 Departmental Representative will prepare WA prior to project start-up. WA will be provided with bid documentation (see Schedule A).
- .2 WA provides detailed inventory, estimated quantities and types of waste materials that will be generated as well as their potential to be reused and/or recycled and project's waste diversion goals and objectives.
- .3 After award of contract, contractor to review WA and confirm that anticipated quantities of waste generated are accurate and goals achievable.
- .4 If after review, contractor determines that indicated quantities or opportunities in WA are not accurate or achievable, contractor to provide written details of discrepancies and revised quantities for areas of concern. Contractor to meet with Departmental & Consultant to review and justify revisions.
- .5 Post on-site WA where contractor and sub-contractors are able to review content.

1.7 WASTE REDUCTION WORKPLAN (WRW)

- .1 Prepare and submit WRW (Schedule B) at least 10 days prior to project start-up.
- .2 WRW identifies strategies to optimize diversion through reduction, reuse, and recycling of materials and comply with applicable regulations, based on information acquired from WA.
- .3 WRW should include but not limited to:

- .1 Applicable regulations.
 - .2 Specific goals for waste reduction, identify existing barriers and develop strategies to overcome them.
 - .3 Destination of materials identified.
 - .4 Deconstruction/disassembly techniques and schedules.
 - .5 Methods to collect, separate, and reduce generated wastes.
 - .6 Location of waste bins on-site.
 - .7 Security of on-site stock piles and waste bins.
 - .8 Protection of personnel, sub-contractors.
 - .9 Clear labelling of storage areas.
 - .10 Training plan for contractor and sub-contractors.
 - .11 Methods to track and report results reliably (Schedule D).
 - .12 Details on materials handling and removal procedures.
 - .13 Recycler and reclaimer requirements.
 - .14 Quantities of materials to be salvaged for reuse or recycled and materials sent to landfill.
 - .15 Requirements for monitoring on-site wastes management activities.
- .4 Structure WRW to prioritize actions and follow 3R's hierarchy, with Reduction as first priority, followed by Reuse, then Recycle.
 - .5 Post WRW or summary where workers at site are able to review content.
 - .6 Monitor and report on waste reduction by documenting total volume and cost of actual waste removed from project (Schedule D).

1.8 COST/REVENUE ANALYSIS WORKPLAN (CRAW)

- .1 Prepare CRAW (see Schedule E) and include the following:
 - .1 Cost of current waste management practices.
 - .2 Implementation cost of waste diversion program.
 - .3 Savings and benefits resulting from waste diversion program.

1.9 WASTE SOURCE SEPARATION PROGRAM (WSSP)

- .1 As part of Waste Reduction Workplan, prepare WSSP prior to project start-up.
- .2 WSSP will detail methodology and planned on-site activities for separation of reusable and recyclable materials from waste intended for landfill.
- .3 Provide list and drawings of locations that will be made available for sorting, collection, handling and storage of anticipated quantities of reusable and recyclable materials.
- .4 Provide sufficient on-site facilities and containers for collection, handling, and storage of anticipated quantities of reusable and recyclable materials.
- .5 Locate containers to facilitate deposit of materials without hindering daily operations.

- .6 Provide training for sub-contractors & workers in handling and separation of materials for reuse and/or recycling.
- .7 Locate separated materials in areas which minimizes material damage.
- .8 Clearly and securely label containers to identify types/conditions of materials accepted and assist sub-contractors & workers in separating materials accordingly.
- .9 Monitor on-site waste management activities by conducting periodic site inspections to verify: state of signage, contamination levels, bin locations and condition, personnel participation, use of waste tracking forms and collection of waybills, receipts and invoices.
- .10 On-site sale of salvaged materials is not permitted unless authorized in writing by Departmental Representative and provided that site safety regulations and security requirements are adhered to.

1.10 USE OF SITE AND FACILITIES

- .1 Execute Work with minimal interference and disturbance to normal use of premises.
- .2 Maintain security measures established by facility provide temporary security measures approved by Departmental Representative.

1.11 WASTE PROCESSING SITES

- .1 Contractor is responsible to research and locate waste diversion resources and service providers. Salvaged materials are to be transported off site to approved and/or authorized recycling facilities or to users of material for recycling.

1.12 QUALITY ASSURANCE

- .1 After award of Contract, a mandatory site examination will be held for this Project for Contractor and/or sub-contractors responsible for construction, renovation demolition/deconstruction waste management.
 - .1 Date, time and location will be arranged by Departmental Representative & Consultant.
- .2 Waste Management Meeting: Waste Management Co-ordinator is to provide an update on status of waste diversion and management activities at each meeting. Written monthly Waste Diversion Report summary to be provided by Waste Management Coordinator (refer to the Waste Diversion Report form in Schedule C and Waste Materials Tracking form in Schedule D).

1.13 STORAGE, HANDLING AND PROTECTION

- .1 Store, materials to be reused, recycled and salvaged in locations as directed by Departmental Representative.
- .2 Unless specified otherwise, materials for removal become Contractor's property.
- .3 Protect, stockpile, store and catalogue salvaged items.

- .4 Separate non-salvageable materials from salvaged items. Transport and deliver non-salvageable items to licensed disposal facility.
- .5 Protect structural components not removed and salvaged materials from movement or damage.
- .6 Support affected structures. If safety of building is endangered, cease operations and immediately notify Departmental Representative & Consultant.
- .7 Protect surface drainage, mechanical and electrical from damage and blockage.
- .8 Provide on-site facilities and containers for collection and storage of reusable and recyclable materials.
- .9 Separate and store materials produced during project in designated areas.
- .10 Prevent contamination of materials to be salvaged and recycled and handle materials in accordance with requirements for acceptance by designated processing facilities.
 - .1 On-site source separation is recommended.
 - .2 Remove co-mingled materials to off site processing facility for separation.
 - .3 Obtain waybills, receipts and/or scale tickets for separated materials removed from site.
 - .4 Materials reused on-site are considered to be diverted from landfill and as such are to be included in all reporting.

1.14 DISPOSAL OF WASTES

- .1 Do not bury rubbish or waste materials.
- .2 Do not dispose of waste, volatile materials, mineral spirits, or oil paint thinner into waterways, storm, or sanitary sewers.
- .3 Keep records of construction waste including:
 - .1 Number and size of bins.
 - .2 Waste type of each bin.
 - .3 Total tonnage generated.
 - .4 Tonnage reused or recycled.
 - .5 Reused or recycled waste destination.
- .4 Remove materials on-site as Work progresses.
- .5 Prepare project summary to verify destination and quantities on a material-by-material basis as identified in the waste audit.

1.15 SCHEDULING

- .1 Co-ordinate Work with other activities at site to ensure timely and orderly progress of Work.

Part 2 Products

2.1 NOT USED

- .1 Not Used.

Part 3 Execution

3.1 APPLICATION

- .1 Do Work in compliance with WRW and WSSP.
- .2 Handle waste materials not reused, salvaged, or recycled in accordance with appropriate regulations and codes.

3.2 CLEANING

- .1 Progress Cleaning: clean in accordance with Section 01 74 11 - Cleaning.
 - .1 Leave Work area clean at end of each day.
- .2 Final Cleaning: upon completion remove surplus materials, rubbish, tools and equipment in accordance with Section 01 74 11 - Cleaning.
- .3 Waste Management: separate waste materials for recycling in accordance with Section 1.9 - 01 74 21 - Construction/Demolition Waste Management and Disposal.
 - .1 Remove recycling containers and bins from site and dispose of materials at appropriate facility.
 - .2 Source separate materials to be reused/recycled into specified sort areas.

3.3 DIVERSION OF MATERIALS

- .1 From following list, separate materials from general waste stream and stockpile in separate piles or containers, as reviewed by the Departmental Representative, and consistent with applicable fire regulations.
 - .1 Mark containers or stockpile areas.
 - .2 Provide instruction on disposal practices.
- .2 On-site sale of salvaged materials is not permitted.

3.4 WASTE DIVERSION REPORT

- .1 At completion of Project, prepare written Waste Diversion Report indicating quantities of materials reused, recycled or disposed of as well as the following:
 - .1 Identify final diversion results and measure success against goals from Waste Reduction Workplan.
 - .2 Compare final quantities/percentages diverted with initial projections in Waste Audit and Waste Reduction Workplan and explain variances.
 - .1 Supporting documentation.

- .2 Waybills and tracking forms.
- .3 Description of issues, resolutions and lessons learned.

3.5 WASTE AUDIT (WA)

.1 Schedule A - Waste Audit (WA)

(1) Material Category	(2) Material Quantity Unit	(3) Estimated Waste %	(4) Total Quantity of Waste (unit)	(5) Generation Point	(6) % Recycled	(7) % Reused
Wood and Plastics Material Description						
Off-cuts						
Warped Pallet Forms						
Plastic Packaging						
Cardboard Packaging						
Other						
Painted Frames						
Glass						
Wood						
Metal						
Other						

3.6 WASTE REDUCTION WORKPLAN (WRW)

.1 Schedule B

(1) Material Category	(2) Person(s) Responsible	(3) Total Quantity of Waste (unit)	(4) Reused Amount (units) Projected	Actual	(5) Recycled Amount (unit) Projected	Actual	(6) Material(s) Destination
Wood and Plastics Material Description							
Chutes							
Warped Pallet							

Forms							
Plastic Packag ing							
Card-board Packag ing							
Other							
Painted Frames							
Glass							
Wood							
Metal							
Other							

3.7 COST/REVENUE ANALYSIS WORKPLAN (CRAW)

.1 Schedule E - Cost/Revenue Analysis Workplan (CRAW)

(1) Material Description	(2) Total Quantity (unit)	(3) Volume (cumulative)	(4) Weight (cumulative)	(5) Disposal Cost/Credit \$(+/-)	(6) Category Sub-Total \$(+/-)
Wood					
Wood Stud					
Plywood					
Slab Regular					
Glazing					\$
		(7) Cost (-) / Revenue (+)			\$

3.8 CANADIAN GOVERNMENTAL DEPARTMENTS CHIEF RESPONSIBILITY FOR THE ENVIRONMENT

.1 Schedule G - Government Chief Responsibility for the Environment:

Province	Address	General Inquires	Fax
Alberta	Alberta Environmental Protection Petroleum Plaza, South Tower 9915 - 108 th Street	403-427-2739	

[illegible]

3.9 SCHEDULES

- .1 Following Schedules are attached to this Specification:
- .1 Waste Audit - Schedule A.
- .2 Waste Reduction Workplan Form - Schedule B.
- .3 Waste Diversion Report Form - Schedule C.
- .4 Waste Materials Tracking Form - Schedule D.
- .5 Cost/Revenue Analysis Workplan - Schedule E.
- .6 Market Research Report - Schedule F (When Available).

END OF SECTION

Part 1 General

1.1 RELATED REQUIREMENTS

- .1 Section 01 33 00.
- .2 Section 01 35 43.
- .3 Section 01 45 00.
- .4 Section 01 74 11.
- .5 Section 01 74 21.
- .6 Section 01 78 00.
- .7 Section 01 35 29.06

1.2 REFERENCES

- .1 Canadian Construction Documents Committee (CCDC)
 - .1 CCDC 2 – (current edition), Stipulated Price Contract.
- .2 Canadian Environmental Protection Act (CEPA)
 - .1 SOR/2008-197, Storage Tank Systems for Petroleum Products and Allied Petroleum Products Regulations.

1.3 ADMINISTRATIVE REQUIREMENTS

- .1 Acceptance of Work Procedures:
 - .1 Contractor's Inspection: Contractor: conduct inspection of Work, identify deficiencies and defects, and repair as required to conform to Contract Documents.
 - .1 Notify Departmental Representative & Consultant in writing of satisfactory completion of Contractor's inspection and submit verification that corrections have been made.
 - .2 Request Departmental Representative and/or Consultant's inspection.
 - .2 Departmental Representative and/or Consultant's Inspection:
 - .1 Departmental Representative, or Consultant and Contractor to inspect Work and identify defects and deficiencies.
 - .2 Contractor to correct Work as directed.
 - .3 Completion Tasks: submit written certificates in English that tasks have been performed as follows:
 - .1 Work: completed and inspected for compliance with Contract Documents.
 - .2 Defects: corrected and deficiencies completed.
 - .3 Equipment and systems: tested, adjusted and fully operational.
 - .4 Certificates required by Utility companies: submitted.
 - .5 Operation of systems: demonstrated to Owner's personnel.

- .6 Work: complete and ready for final inspection.
- .4 Final Inspection:
 - .1 When completion tasks are done, request final inspection of Work by Departmental Representative & Consultant, and Contractor.
 - .2 When Work is incomplete according to Owner, Departmental Representative, and/or Consultant, complete outstanding items and request re-inspection.
- .5 Declaration of Substantial Performance: when Departmental Representative and/or Consultant considers deficiencies and defects corrected and requirements of Contract substantially performed, make application for Certificate of Substantial Performance.
- .6 Commencement of Lien and Warranty Periods: date of Owner's acceptance of submitted declaration of Substantial Performance to be date for commencement for warranty period and commencement of lien period unless required otherwise by lien statute of Place of Work.
- .7 Final Payment:
 - .1 When Departmental Representative and/or Consultant considers final deficiencies and defects corrected and requirements of Contract met, make application for final payment.
 - .2 Refer to CCDC 2: when Work is deemed incomplete by Departmental Representative, or Consultant, complete outstanding items and request re-inspection.
- .8 Payment of Holdback: after issuance of Certificate of Substantial Performance of Work, submit application for payment of holdback amount in accordance with contractual agreement.

1.4 FINAL CLEANING

- .1 Clean in accordance with Section 01 74 11 - Cleaning.
 - .1 Remove surplus materials, excess materials, rubbish, tools and equipment.
- .2 Waste Management: separate waste materials for recycling in accordance with Section 01 74 21 - Construction/Demolition Waste Management and Disposal.

Part 2 Products

2.1 NOT USED

- .1 Not Used.

Part 3 Execution

3.1 NOT USED

- .1 Not Used.

END OF SECTION

Part 1 General

1.1 RELATED REQUIREMENTS

- .1 Section 01 33 00.
- .2 Section 01 35 43.
- .3 Section 01 45 00.
- .4 Section 01 74 11.
- .5 Section 01 74 21.
- .6 Section 01 77 00.
- .7 Section 01 35 29.06

1.2 REFERENCES

- .1 Canadian Environmental Protection Act (CEPA)
 - .1 SOR/2008-197, Storage Tank Systems for Petroleum Products and Allied Petroleum Products Regulations.

1.3 ADMINISTRATIVE REQUIREMENTS

- .1 Pre-warranty Meeting:
 - .1 Convene meeting one week prior to contract completion with contractor's representative, Departmental Representative and Consultant to:
 - .1 Verify Project requirements.
 - .2 Review warranty requirements and manufacturer's installation instructions.
 - .2 Departmental Representative to establish communication procedures for:
 - .1 Notifying construction warranty defects.
 - .2 Determine priorities for type of defects.
 - .3 Determine reasonable response time.
 - .3 Contact information for bonded and licensed company for warranty work action: provide name, telephone number and address of company authorized for construction warranty work action.
 - .4 Ensure contact is located within local service area of warranted construction, is continuously available, and is responsive to inquiries for warranty work action.

1.4 ACTION AND INFORMATIONAL SUBMITTALS

- .1 Provide submittals in accordance with Section 01 33 00 - Submittal Procedures.
- .2 Two weeks prior to Substantial Performance of the Work, submit to the Departmental Representative and Consultant, one digital final copy of operating and maintenance manuals in English.

- .3 Provide spare parts, maintenance materials and special tools of same quality and manufacture as products provided in Work.
- .4 Provide evidence, if requested, for type, source and quality of products supplied.

1.5 FORMAT

- .1 Organize data as instructional manual.
- .2 Binders (Provide 2 hard copies): vinyl, hard covered, 3 'D' ring, loose leaf 219 x 279 mm with spine and face pockets.
- .3 When multiple binders are used correlate data into related consistent groupings.
 - .1 Identify contents of each binder on spine.
- .4 Cover: identify each binder with type or printed title 'Project Record Documents'; list title of project and identify subject matter of contents.
- .5 Arrange content by systems, under Section numbers and sequence of Table of Contents.
- .6 Provide tabbed fly leaf for each separate product and system, with typed description of product and major component parts of equipment.
- .7 Text: manufacturer's printed data, or typewritten data.
- .8 Drawings: provide with reinforced punched binder tab.
 - .1 Bind in with text; fold larger drawings to size of text pages.
- .9 Provide scaled CAD files in dwg format on memory stick.

1.6 CONTENTS - PROJECT RECORD DOCUMENTS

- .1 Table of Contents for Each Volume: provide title of project;
 - .1 Date of submission; names.
 - .2 Addresses, and telephone numbers of Consultant and Contractor with name of responsible parties.
 - .3 Schedule of products and systems, indexed to content of volume.
- .2 For each product or system:
 - .1 List names, addresses and telephone numbers of subcontractors and suppliers, including local source of supplies and replacement parts.
- .3 Product Data: mark each sheet to identify specific products and component parts, and data applicable to installation; delete inapplicable information.
- .4 Drawings: supplement product data to illustrate relations of component parts of equipment and systems, to show control and flow diagrams.
- .5 Typewritten Text: as required to supplement product data.
 - .1 Provide logical sequence of instructions for each procedure, incorporating manufacturer's instructions specified in Section 01 45 00 - Quality Control.

1.7 AS -BUILT DOCUMENTS AND SAMPLES

- .1 Maintain, in addition to requirements in General Conditions, at site for Departmental Representative and Consultant one record copy of:
 - .1 Contract Drawings.
 - .2 Specifications.
 - .3 Addenda.
 - .4 Change Orders and other modifications to Contract.
 - .5 Reviewed shop drawings, product data, and samples.
 - .6 Field test records.
 - .7 Inspection certificates.
 - .8 Manufacturer's certificates.
- .2 Store record documents and samples in field office apart from documents used for construction.
 - .1 Provide files, racks, and secure storage.
- .3 Label record documents and file in accordance with Section number listings in List of Contents of this Project Manual.
 - .1 Label each document "PROJECT RECORD" in neat, large, printed letters.
- .4 Maintain record documents in clean, dry and legible condition.
 - .1 Do not use record documents for construction purposes.
- .5 Keep record documents and samples available for inspection by Departmental Representative and/or Consultant.

1.8 RECORDING INFORMATION ON PROJECT RECORD DOCUMENTS

- .1 Record information on set of black line opaque drawings, provided by Consultant.
- .2 Use felt tip marking pens, maintaining separate colours for each major system, for recording information.
- .3 Record information concurrently with construction progress.
 - .1 Do not conceal Work until required information is recorded.
- .4 Contract Drawings and shop drawings: mark each item to record actual construction, including:
 - .1 Measured depths of elements of foundation in relation to finish first floor datum.
 - .2 Measured horizontal and vertical locations of underground utilities and appurtenances, referenced to permanent surface improvements.
 - .3 Measured locations of internal utilities and appurtenances, referenced to visible and accessible features of construction.
 - .4 Field changes of dimension and detail.
 - .5 Changes made by change orders.

- .6 Details not on original Contract Drawings.
- .7 References to related shop drawings and modifications.
- .5 Specifications: mark each item to record actual construction, including:
 - .1 Manufacturer, trade name, and catalogue number of each product actually installed, particularly optional items and substitute items.
 - .2 Changes made by Addenda and change orders.
- .6 Other Documents: maintain manufacturer's certifications, inspection certifications, field test records, required by individual specifications sections.
- .7 Provide digital photos, if requested, for site records.

1.9 FINAL SURVEY

- .1 Submit final site survey certificate, certifying that elevations and locations of completed Work are in conformance, or non-conformance with Contract Documents.

1.10 EQUIPMENT AND SYSTEMS

- .1 For each item of equipment and each system include description of unit or system, and component parts.
 - .1 Give function, normal operation characteristics and limiting conditions.
 - .2 Include performance curves, with engineering data and tests, and complete nomenclature and commercial number of replaceable parts.
- .2 Panel board circuit directories: provide electrical service characteristics, controls, and communications.
- .3 Include installed colour coded wiring diagrams.
- .4 Operating Procedures: include start-up, break-in, and routine normal operating instructions and sequences.
 - .1 Include regulation, control, stopping, shut-down, and emergency instructions.
 - .2 Include summer, winter, and any special operating instructions.
- .5 Maintenance Requirements: include routine procedures and guide for trouble-shooting; disassembly, repair, and reassembly instructions; and alignment, adjusting, balancing, and checking instructions.
- .6 Include manufacturer's printed operation and maintenance instructions.
- .7 Provide original manufacturer's parts list, illustrations, assembly drawings, and diagrams required for maintenance.
- .8 Provide installed control diagrams by controls manufacturer.
- .9 Provide list of original manufacturer's spare parts, current prices, and recommended quantities to be maintained in storage.
- .10 Additional requirements: as specified in individual specification sections.

1.11 MATERIALS AND FINISHES

- .1 Building products, applied materials, and finishes: include product data, with catalogue number, size, composition, and colour and texture designations.
- .2 Instructions for cleaning agents and methods, precautions against detrimental agents and methods, and recommended schedule for cleaning and maintenance.
- .3 Moisture-protection and weather-exposed products: include manufacturer's recommendations for cleaning agents and methods, precautions against detrimental agents and methods, and recommended schedule for cleaning and maintenance.
- .4 Additional requirements: as specified in individual specifications sections.

1.12 MAINTENANCE MATERIALS

- .1 Spare Parts:
 - .1 Provide spare parts, in quantities specified in individual specification sections.
 - .2 Provide items of same manufacture and quality as items in Work.
 - .3 Deliver to location as directed; place and store.
 - .4 Receive and catalogue items.
 - .1 Submit inventory listing to Departmental Representative and Consultant.
 - .2 Include approved listings in Maintenance Manual.
 - .5 Obtain receipt for delivered products and submit prior to final payment.

1.13 DELIVERY, STORAGE AND HANDLING

- .1 Store spare parts, maintenance materials, and special tools in manner to prevent damage or deterioration.
- .2 Store in original and undamaged condition with manufacturer's seal and labels intact.
- .3 Store components subject to damage from weather in weatherproof enclosures.
- .4 Store paints and freezable materials in a heated and ventilated room.
- .5 Remove and replace damaged products at own expense and for review by Departmental Representative and/or Consultant.

1.14 WARRANTIES AND BONDS

- .1 Develop warranty management plan to contain information relevant to Warranties.
- .2 Submit warranty management plan, 30 days before planned pre-warranty conference, to Departmental Representative and Consultant approval.
- .3 Warranty management plan to include required actions and documents to assure that Departmental Representative and Consultant receives warranties to which it is entitled.
- .4 Provide plan in narrative form and contain sufficient detail to make it suitable for use by future maintenance and repair personnel.

-
- .5 Submit, warranty information made available during construction phase, to Departmental Representative and Consultant for approval prior to each monthly pay estimate.
 - .6 Assemble approved information in binder, submit upon acceptance of work and organize binder as follows:
 - .1 Separate each warranty or bond with index tab sheets keyed to Table of Contents listing.
 - .2 List subcontractor, supplier, and manufacturer, with name, address, and telephone number of responsible principal.
 - .3 Obtain warranties and bonds, executed in duplicate by subcontractors, suppliers, and manufacturers, within ten days after completion of applicable item of work.
 - .4 Verify that documents are in proper form, contain full information, and are notarized.
 - .5 Co-execute submittals when required.
 - .6 Retain warranties and bonds until time specified for submittal.
 - .7 Except for items put into use with Owner's permission, leave date of beginning of time of warranty until Date of Substantial Performance is determined.
 - .8 Conduct joint 9 month warranty inspection, measured from time of acceptance, by Departmental Representative and Consultant.
 - .9 Include information contained in warranty management plan as follows:
 - .1 Roles and responsibilities of personnel associated with warranty process, including points of contact and telephone numbers within the organizations of Contractors, subcontractors, manufacturers or suppliers involved.
 - .2 Listing and status of delivery of Certificates of Warranty for extended warranty items, to include transformers, commissioned systems, surge suppression systems.
 - .3 Provide list for each warranted equipment, item, feature of construction or system indicating:
 - .1 Name of item.
 - .2 Model and serial numbers.
 - .3 Location where installed.
 - .4 Name and phone numbers of manufacturers or suppliers.
 - .5 Names, addresses and telephone numbers of sources of spare parts.
 - .6 Warranties and terms of warranty: include one-year overall warranty of construction. Indicate items that have extended warranties and show separate warranty expiration dates.
 - .7 Cross-reference to warranty certificates as applicable.
 - .8 Starting point and duration of warranty period.
 - .9 Summary of maintenance procedures required to continue warranty in force.

- .10 Cross-Reference to specific pertinent Operation and Maintenance manuals.
- .11 Organization, names and phone numbers of persons to call for warranty service.
- .12 Typical response time and repair time expected for various warranted equipment.
- .4 Contractor's plans for attendance at 9 month post-construction warranty inspections.
- .5 Procedure and status of tagging of equipment covered by extended warranties.
- .6 Post copies of instructions near selected pieces of equipment where operation is critical for warranty and/or safety reasons.
- .10 Respond in timely manner to oral or written notification of required construction warranty repair work.
- .11 Written verification to follow oral instructions.
- .1 Failure to respond will be cause for the Departmental Representative and/or Consultant to proceed with action against Contractor.

1.15 WARRANTY TAGS

- .1 Tag, at time of installation, each warranted item. Provide durable, oil and water resistant tag approved by Departmental Representative and/or Consultant.
- .2 Attach tags with copper wire and spray with waterproof silicone coating.
- .3 Leave date of acceptance until project is accepted for occupancy.
- .4 Indicate following information on tag:
 - .1 Type of product/material.
 - .2 Model number.
 - .3 Serial number.
 - .4 Contract number.
 - .5 Warranty period.
 - .6 Inspector's signature.
 - .7 Construction Contractor.

Part 2 Products

2.1 NOT USED

- .1 Not Used.

Part 3 Execution

3.1 NOT USED

.1 Not Used.

END OF SECTION